Exercise J-M

1. Let $y = y(x)$ be the solution of the differential equation $x^3\,dy + (xy - 1)dx = 0$, $x > 0$, $y\left(\frac{1}{2}\right) = 3 - e$. Then $y(1)$ is equal to : **[JEE(Main)Jan.2023]**

माना अवकल समीकरण $x^3\,dy + (xy - 1)dx = 0$, $x > 0$, $y\left(\frac{1}{2}\right) = 3 - e$ का हल $y = y(x)$ है। तो $y(1)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $2 - e$ (B) e (C) 1 (D) 3

Ans. C

Sol. $\frac{dy}{dx} = \frac{1-xy}{x^3} = \frac{1}{x^3} - \frac{y}{x^2}$

$\frac{dy}{dx} + \frac{y}{x^2} = \frac{1}{x^3}$

If $= e^{\int \frac{1}{x^2}dx} = e^{-\frac{1}{x}}$

$y \cdot e^{-\frac{1}{x}} = \int e^{-\frac{1}{x}} \cdot \frac{1}{x^3}dx$ (put $-\frac{1}{x} = t$)

$y = \frac{1}{x} + 1 + Ce^{\frac{1}{x}}$

Where C is constant

Put $x = \frac{1}{2}$

$3 - e = 2 + 1 + Ce^2$

$C = -\frac{1}{e}$

$y(1) = 1$

2. Let $y = y(x)$ be the solution of the differential equation $(x^2 - 3y^2)\,dx + 3xy\,dy = 0$, $y(1) = 1$. Then $6\,y^2(e)$ is equal to : **[JEE(Main)Jan.2023]**

माना अवकल समीकरण $(x^2 - 3y^2)\,dx + 3xy\,dy = 0$, $y(1) = 1$ का हल $y = y(x)$ है। तब $6\,y^2(e)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $3e^2$ (B) e^2 (C) $\frac{3}{2}e^2$ (D) $2e^2$

Ans. D

Sol. D.E. $\rightarrow \frac{2ydy}{dx} - \frac{2y^2}{x} = \frac{-2x}{3}$

Let $y^2 = t \Rightarrow 2y\frac{dy}{dx} = \frac{dt}{dx}$

D.E. $\rightarrow \frac{dt}{dx} - \frac{2t}{x} = -\frac{2x}{3}$

I.F. $= e^{\int -\frac{2}{x}dx} = e^{-2en|x|} = e^{tn\frac{1}{x^2}} = \frac{1}{x^2}$

Solution $\Rightarrow t \cdot \frac{1}{x^2} = \int \frac{1}{x^2}\left(\frac{-2x}{3}\right)dx + c \Rightarrow \frac{y^2}{x^2} = \frac{-2}{3}\ell n|x| + c$

$\therefore x = 1, y = 1 \Rightarrow 1 = 0 + c \Rightarrow c = 1$

$\Rightarrow \frac{y^2}{x^2} = -\frac{2}{3}\ell n|x| + 1$

$x = e \Rightarrow \frac{y^2(e)}{e^2} = \frac{-2}{3} + 1 \Rightarrow y^2(e) = \frac{e^2}{3} \Rightarrow 6y^2(e) = 2e^2$

3. Let y = y(x) be the solution curve of the differential equation $\frac{dy}{dx} = \frac{y}{x}\left(1 + xy^2(1 + \log_e x)\right)$, x > 0, y(1) =3. Then $\frac{y^2(x)}{9}$ is equal to : **[JEE(Main)Jan.2023]**

माना अवकल समीकरण $\frac{dy}{dx} = \frac{y}{x}\left(1 + xy^2(1 + \log_e x)\right)$, x > 0, y(1) =3 का हल y = y(x) है। तो $\frac{y^2(x)}{9}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{x^2}{7 - 3x^2\left(2 + \log_e x^2\right)}$

(B) $\frac{x^2}{2x^3\left(2 + \log_e x^3\right) - 3}$

(C) $\frac{x^2}{3x^3\left(1 + \log_e x^2\right) - 2}$

(D) $\frac{x^2}{5 - 2x^3\left(2 + \log_e x^3\right)}$

Ans. D

Sol. $\frac{dy}{dx} = \frac{y}{x}\cdot\left(1 + xy^2(1 + \log_e x)\right), x > 0$

$\frac{dy}{dx} = \frac{y}{x} + y^3\cdot\left(1 + \log_e x\right)$

$\frac{1}{y^3}\cdot\frac{dy}{dx} = \frac{1}{x\cdot y^2} + \left(1 + \log_e x\right)$

$\frac{1}{y^3}\cdot\frac{dy}{dx} - \frac{1}{x}\cdot\frac{1}{y^2} = \left(1 + \ln x\right)$

Let $\frac{1}{y^2}=t$

$-\frac{2}{y^3}\cdot\frac{dy}{dx}=\frac{dt}{dx}$

$\therefore -\frac{1}{2}\frac{dt}{dx}-\frac{t}{x}=(1+\ln x)$

$\frac{dt}{dx}+\frac{2t}{x}=-2(1+\ln x)$

if $e^{\int\frac{2}{x}dx}=x^2$

$\therefore tx^2=\int -2(1+\ln x)x^2\cdot dx$

$tx^2=-2\left[(1+\ln x)\frac{x^3}{3}-\int\frac{x^2}{3}dx\right]+C$

$\frac{x^2}{y^2}=-2\left[\frac{x^3}{3}(1+\ln x)-\frac{x^3}{9}\right]+C$ ...(1)

$\because y(1)=3$

$\frac{1}{9}=-2\left[\frac{1}{3}(1+0)-\frac{1}{9}\right]+C$

$c=\frac{5}{9}$

put $c=\frac{5}{9}$

$\Rightarrow\frac{y^2(x)}{9}=\frac{x^2}{5-2x^3\left(2+\log_e x^3\right)}$

4. Let $y = y(t)$ be a solution of the differential equation

$\frac{dy}{dt}+\alpha y=\gamma e^{-\beta t}$

where, $\alpha>0,\beta>0$ and $\gamma>0$. Then $\lim\limits_{t\to\infty}y(t)$ **[JEE(Main)Jan.2023]**

(A) is -1 (B) is 0 (C) does not exist (D) is 1

माना अवकल समीकरण

$\frac{dy}{dt}+\alpha y=\gamma e^{-\beta t}$

जहाँ $\alpha > 0, \beta > 0$ तथा $\gamma > 0$ है, का हल $y = y(t)$ है। तो $\lim_{t\to\infty} y(t)$ **[JEE(Main)Jan.2023]**

(A) –1 है (B) 0 है (C) का अस्तित्व नहीं है (D) 1 है

Ans. B

Sol. I.F. $= e^{\int adt} = e^{at}$

Solution $y.e^{at} = \int e^{at} \cdot \gamma \cdot e^{-at} dt + C$

$\Rightarrow ye^{\alpha t} = \gamma \int e^{(\alpha-\beta)t} dt + c$

$\Rightarrow y(t) \cdot e^{\alpha t} = \gamma \frac{e^{(\alpha-\beta)t}}{(\alpha-\beta)} + c$

$\Rightarrow y(t) = \gamma \frac{e^{-\beta t}}{(\alpha-\beta)} + ce^{-\alpha t}$

$\lim_{t\to\infty} y(t) = \lim_{t\to\infty} \left(\frac{\gamma \cdot e^{-\beta t}}{(\alpha-\beta)} - c \cdot e^{-at} \right)$

$= 0 - 0 = 0$

5. Let $y = f(x)$ be the solution of the differential equation $y(x + 1)dx - x^2dy = 0, y(1) = e$. Then $\lim_{x\to 0^+} f(x)$ is equal to: **[JEE(Main)Jan.2023]**

माना $y = f(x)$ अवकल समीकरण $y(x + 1)dx - x^2dy = 0, y(1) = e$ का हल है। तब $\lim_{x\to 0^+} f(x)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) e^2 (B) $\frac{1}{e}$ (C) 0 (D) $\frac{1}{e^2}$

Ans. C

Sol. $\frac{x+1}{x^2} dx = \frac{dy}{y}$

$\ln x - \frac{1}{x} = \ln y + c$

$(1, e)$

$c = -2$

$\ln x - \frac{1}{x} = \ln y - 2$

$y = e^{\ln x - \frac{1}{x} + 2}$

$= \lim_{x\to 0^+} e^{\ln x - \frac{1}{x} + 2}$

$= e^{-\infty}$

$= 0$

6. Let $y = y(x)$ be the solution of the differential equation $x \log_e x \frac{dy}{dx} + y = x^2 \log_e x$, $(x > 1)$. If $y(2) = 2$, then $y(e)$ is equal to : **[JEE(Main)Jan.2023]**

माना अवकल समीकरण $x \log_e x \frac{dy}{dx} + y = x^2 \log_e x$, $(x > 1)$ का हल $y = y(x)$ है। यदि $y(2) = 2$ है, तो $y(e)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{1+e^2}{2}$ (B) $\frac{1+e^2}{4}$ (C) $\frac{4+e^2}{4}$ (D) $\frac{2+e^2}{2}$

Ans. C

Sol. $x \log_e x \frac{dy}{dx} + y = x^2 \log_e x, (x > 1)$

$\Rightarrow \frac{dy}{dx} + \frac{y}{x \ln x} = x$

Linear differential equation

I.F. = $e^{\int \frac{1}{x \ln x} dx} = |\ln x|$

$\therefore$ Solution of differential equation

$y|\ln x| = \int x|\ln x| dx$

$\left|\ln x^2\right| \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx$

$\Rightarrow y|\ln x| = |\ln x|\left(\frac{x^2}{2}\right) - \frac{x^2}{4} + c$

For constant

$y(2) = 2 \Rightarrow c = 1$

So, $y(x) = \frac{x^2}{2} - \frac{x^2}{4|\ln x|} + \frac{1}{|\ln x|}$

Hence, $y(e) = \frac{e^2}{2} - \frac{e^2}{4} + 1 = 1 + \frac{e^2}{4}$

7. Let the solution curve $y = y(x)$ of the differential equation

$\frac{dy}{dx} - \frac{3x^5 \tan^{-1}(x^3)}{(1+x^6)^{3/2}} y = 2x \exp\left\{\frac{x^3 - \tan^{-1} x^3}{\sqrt{(1+x^6)}}\right\}$ pass through the origin. Then $y(1)$ is equal to :

[JEE(Main)Jan.2023]

माना अवकल समीकरण $\frac{dy}{dx}-\frac{3x^5\tan^{-1}(x^3)}{(1+x^6)^{3/2}}y=2x\exp\left\{\frac{x^3-\tan^{-1}x^3}{\sqrt{(1+x^6)}}\right\}$ का हल वक्र $y = y(x)$ मूल बिंदु से होकर जाता है।

r ksy(1) बराबर है : **[JEE(Main)Jan.2023]**

(A) $\exp\left(\frac{\pi-4}{4\sqrt{2}}\right)$ (B) $\exp\left(\frac{1-\pi}{4\sqrt{2}}\right)$ (C) $\exp\left(\frac{4+\pi}{4\sqrt{2}}\right)$ (D) $\exp\left(\frac{4-\pi}{4\sqrt{2}}\right)$

Ans. D

Sol. $\frac{dy}{dx}-\frac{3x^5\tan^{-1}x^3}{\left(1+x^6\right)^{3/2}}y=2x\exp\left\{\frac{x^3-\tan^{-1}x^3}{\sqrt{1+x^6}}\right\}$

$\text{I.f.}=e^{-\int\frac{3x^5\tan^{-1}x^3}{\left(1+x^6\right)^{3/2}}dx}$

$=e^{-\int\frac{3x^2.x^3\tan^{-1}x^3}{\left(1+x^6\right)^{3/2}}dx}$

Let $\tan^{-1}x^3=t\Rightarrow\frac{3x^2}{1+x^6}dx=dt$

$=e^{-\int\frac{x^3}{\sqrt{1+x^6}}.\frac{3x^2\tan^{-1}x^3}{\left(1+x^6\right)}dx}$

$=e^{-\int\frac{\tan t}{\sec t}.t.dt}$

$=e^{-\int t\sin t.dt}=e^{(t\cos t-\sin t)}$

$=e^{\left(\frac{\tan^{-1}x^3}{\sqrt{1+x^6}}-\frac{x^3}{\sqrt{1+x^6}}\right)}$

Hence solutions of the differential equation is

$y.e^{\frac{\tan^{-1}x^3}{\sqrt{1+x^6}}-\frac{x^3}{\sqrt{1+x^6}}}=\int 2\times e^{\frac{x^3-\tan^{-1}x^3}{\sqrt{1+x^6}}}.e^{\frac{\tan^{-1}x^3-x^3}{\sqrt{1+x^6}}}.dx+C$

$= x^2 + C$

Passes through $(0, 0)\Rightarrow C = 0$

Hence $y(1)=e^{\frac{1}{\sqrt{2}}\left(1-\frac{\pi}{4}\right)}$

8. The solution of the differential equation $\frac{dy}{dx}=-\left(\frac{x^2+3y^2}{3x^2+y^2}\right)$, $y(1) = 0$ is : **[JEE(Main)Jan.2023]**

अवकल समीकरण $\frac{dy}{dx}=-\left(\frac{x^2+3y^2}{3x^2+y^2}\right)$, $y(1)=0$ का हल है : **[JEE(Main)Jan.2023]**

(A) $\log_e|x+y|-\frac{xy}{(x+y)^2}=0$ (B) $\log_e|x+y|+\frac{xy}{(x+y)^2}=0$

(C) $\log_e|x+y|+\frac{2xy}{(x+y)^2}=0$ (D) $\log_e|x+y|-\frac{2xy}{(x+y)^2}=0$

Ans. C

Sol. Let $y=vx$

$$v+x\frac{dv}{dx}=-\left(\frac{1+3v^2}{3+v^2}\right)$$

$$\Rightarrow \frac{v^2+3}{(v+1)^3}dv+\frac{1}{x}dx=0$$

$$\Rightarrow \int\frac{4}{(v+1)^3}dv+\int\frac{1}{v+1}dv-\int\frac{2}{(v+1)^2}dv+\int\frac{1}{x}dx=0$$

$$\Rightarrow \frac{-2}{(v+1)^2}+\ln|v+1|+\frac{2}{v+1}+\ln|x|=c$$

$$\Rightarrow \frac{-2x^2}{(x+y)^2}+\ln\left|\frac{x+y}{x}\right|+\frac{2x}{x+y}+\ln|x|=c$$

$$\Rightarrow \frac{2xy}{(x+y)^2}+\ln|x+y|=c$$

$\because y(1)=0$

$\Rightarrow c=0$

$$\Rightarrow \frac{2xy}{(x+y)^2}+\ln|x+y|=0$$

9. Let a differentiable function f satisfy $f(x)+\int_3^x\frac{f(t)}{t}dt=\sqrt{x+1}, x\geq 3$. Then $12f(8)$ is equal to :

[JEE(Main)Jan.2023]

माना एक अवकलनीय फलन f, $f(x)+\int_3^x\frac{f(t)}{t}dt=\sqrt{x+1}, x\geq 3$ को संतुष्ट करता है। तो $12f(8)$ बराबर है :

[JEE(Main)Jan.2023]

(A) 1 (B) 34 (C) 19 (D) 17

Ans. D

Sol. $f'(x)+\frac{f(x)}{x}=\frac{1}{2\sqrt{x+1}}$

$\frac{dy}{dx}+\frac{y}{x}=\frac{1}{2\sqrt{x+1}}$

I.F. $=e^{\int\frac{1}{x}dx}=x$

$y.x=\int x\frac{1}{2\sqrt{x+1}}dx+c$ put $x+1=t^2$, $dx=2tdt$

$y.x=\frac{1}{2}\int\frac{\left(t^2-1\right)2tdt}{t}+c$

$y.x=\frac{1}{2}\left(\frac{2}{3}t^3-2t\right)+c$

$y.x=\frac{\left(\sqrt{x+1}\right)^3}{3}-\sqrt{x+1}+c$

now $f(3)=2$

$6=\frac{(2)^3}{3}-2+c$

$8-\frac{8}{3}=c$

$\frac{16}{3}=c$

$\therefore f(x)=\frac{1}{x}\left(\frac{\left(\sqrt{x+1}\right)^3}{3}-\left(\sqrt{x+1}\right)+\frac{16}{3}\right)$

$\therefore 12f(8)=\frac{12}{8}\left(\frac{27}{3}-3+\frac{16}{3}\right)$

$=\frac{3}{2}\left(6+\frac{16}{3}\right)=17$

10. Let y = y(x) be the solution of the differential equation $(3y^2 - 5x^2)y\,dx + 2x(x^2 - y^2)\,dy = 0$ such that y(1) = 1. Then $|(y(2))^3 - 12y(2)|$ is equal to : **[JEE(Main)Jan.2023]**

माना अवकल समीकरण $(3y^2 - 5x^2)y\,dx + 2x(x^2 - y^2)\,dy = 0$, y(1) = 1 का हल y = y(x) है तो $|(y(2))^3 - 12y(2)|$ बराबर है: **[JEE(Main)Jan.2023]**

(A) $32\sqrt{2}$ (B) $16\sqrt{2}$ (C) 32 (D) 64

Ans. A

Sol. $(3y2 - 5x^2)\, ydx + 2x\,(x^2 - y^2)\, dy = 0$

$$\Rightarrow \frac{dy}{dx} = \frac{y\left(5x^2 - 3y^2\right)}{2x\left(x^2 - y^2\right)}$$

This is homogeneous

Let $y = tx \Rightarrow \frac{dy}{dx} = t + x\frac{dt}{dx}$

$$t + x\frac{dt}{dx} = \frac{(tx)\left(5x^2 - 3t^2x^2\right)}{2x\left(x^2 - t^2x^2\right)} = \frac{t\left(5 - 3t^2\right)}{2\left(1 - t^2\right)}$$

$$x\frac{dt}{dx} = \frac{5t - 3t^3}{2 - 2t^2} - t = \frac{3t - t^3}{2 - 2t^2}$$

$$\int \frac{\left(1 - t^2\right)}{3t - t^3} dt = \int \frac{1}{x} dx$$

Let $3t - t^3 = z \Rightarrow = 3(1 - t^2)dt = dz$

11. If $y = y(x)$ is the solution curve of the differential equation $\frac{dy}{dx} + y\tan x = x\sec x$, $0 \le x \le \frac{\pi}{3}$, $y(0) = 1$, then $y\left(\frac{\pi}{6}\right)$ is equal to : **[JEE(Main)Jan.2023]**

यदि अवकल समीकरण $\frac{dy}{dx} + y\tan x = x\sec x$, $0 \le x \le \frac{\pi}{3}$, $y(0) = 1$ का हल वक्र $y = y(x)$ है, तो $y\left(\frac{\pi}{6}\right)$ का मान है : **[JEE(Main)Jan.2023]**

(A) $\frac{\pi}{12} - \frac{\sqrt{3}}{2}\log_e\left(\frac{2\sqrt{3}}{e}\right)$ (B) $\frac{\pi}{12} + \frac{\sqrt{3}}{2}\log_e\left(\frac{2\sqrt{3}}{e}\right)$

(C) $\frac{\pi}{12} - \frac{\sqrt{3}}{2}\log_e\left(\frac{2}{e\sqrt{3}}\right)$ (D) $\frac{\pi}{12} + \frac{\sqrt{3}}{2}\log_e\left(\frac{2}{e\sqrt{3}}\right)$

Ans. C

Sol. Here I.F. = sec x

Then solution of D.E.

$y(\sec x) = x\tan x - \ln(\sec x) + c$

Given $y(0) = 1 \Rightarrow c = 1$

$\therefore\ y(\sec x) = x\tan x - \ln(\sec x) + 1$

At $x = \frac{\pi}{6}$, $y = \frac{\pi}{12} + \frac{\sqrt{3}}{2}\ln\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}$

$$y = \frac{\pi}{12} + \frac{\sqrt{3}}{2}\left(\log\frac{\sqrt{3}}{2} + \log e\right)$$

$$= \frac{\pi}{12} - \frac{\sqrt{3}}{2}\log\left(\frac{2}{e\sqrt{3}}\right)$$

12. Let $\alpha x = \exp\left(x^{\beta}y^{\gamma}\right)$ be the solution of the differential equation $2x^2y\,dy - (1-xy^2)dx = 0, x > 0,\ y(2) = \sqrt{\log_e 2}$. Then $\alpha + \beta - \gamma$ equals : **[JEE(Main)Jan.2023]**

माना अवकल समीकरण $2x^2y\,dy - (1-xy^2)dx = 0, x > 0,\ y(2) = \sqrt{\log_e 2}$ का हल $\alpha x = \exp\left(x^{\beta}y^{\gamma}\right)$ है। तो $\alpha + \beta - \gamma$: **[JEE(Main)Jan.2023]**

(A) -1 (B) 1 (C) 0 (D) 3

Ans. B

Sol. $\alpha x = e^{x^{\beta}.y^{\gamma}}$

$$2x^2y\frac{dy}{dx} = 1 - x.y^2 \qquad y^2 = t$$

$$x^2\frac{dt}{dx} = 1 - xt$$

$$\frac{dt}{dx} + \frac{t}{x} = \frac{1}{x^2} \qquad \text{I.F.} = e^{\ln x} = x$$

$$t(x) = \int \frac{1}{x^2}.x\,dx$$

$y^2 . x = \ln x + C$

$\therefore 2.\ln 2 = \ln 2 + C$

$\therefore C = \ln 2$

Hence, $xy^2 = \ln 2x$

$\therefore 2x = e^{x.y^2}$

Hence $\alpha = 2, \beta = 1, \gamma = 2$

13. Let $f: R \to R$ be differentiable function such that $f'(x) + f(x) = \int_0^2 f(t)\,dt$. If $f(0) = e^{-2}$, then $2f(0) - f(2)$ is equal to _____. **[JEE(Main)Jan.2023]**

माना $f: R \to R$ एक अवकलनीय फलन है, जिसके लिए ताकि $f'(x) + f(x) = \int_0^2 f(t)\,dt$ है। यदि $f(0) = e^{-2}$ है, तो $2f(0) - f(2)$ का मान है। **[JEE(Main)Jan.2023]**

Ans. 1

Sol. $\frac{dy}{dx} + y = k$

$y \cdot e^x = k \cdot e^x + c$

f(0) = e^{-2}

$\Rightarrow c = e^{-2} - k$

$\therefore y = k + (e^{-2} - k)\, e^{-x}$

now $k = \int_0^2 \left(k + \left(e^{-2} - k\right)e^{-x}\right)dx$

$\Rightarrow k = e^{-2} - 1$

$\therefore y = (e^{-2} - 1) + e^{-x}$

$f(2) = 2e^{-2} - 1,\ f(0) = e^{-2}$

$2f(0) - f(2) = 1$

14. If the solution curve f(x, y)=0 of the differential equation $\left(1 + \log_e x\right)\frac{dx}{dy} - x \log_e x = e^y, x > 0$, passes through the points (1, 0) and (α, 2) then α^α is equal to : **[JEE(Main)April,2023]**

यदि अवकल समीकरण $\left(1 + \log_e x\right)\frac{dx}{dy} - x \log_e x = e^y, x > 0$ का हल वक्र f(x, y) = 0 बिंदुओं (1, 0) तथा (α, 2) से होकर t kr kgs$r ks$\alpha^\alpha$ बराबर है : **[JEE(Main)April,2023]**

(A) e^{2e^2} (B) e^{e^2} (C) $e^{2e^{\sqrt{2}}}$ (D) $e^{\sqrt{2}e^2}$

Ans. A

Sol. $(1 + \ln x)\frac{dx}{dy} - x \ln x = e^y$

Let $x \ln x = t$

$(1 + \ln x)\frac{dx}{dy} = \frac{dt}{dy}$

$\frac{dt}{dy} - t = e^y$

If $= e^{\int -dy} = e^{-y}$

$t \cdot e^{-y} = \int e^y e^{-y} dy + c$

$te^{-y} = y + c$

$x \ln x e^{-y} = y + c$

$x \ln x = ye^y + ce^y$

$(1,0)0 = C$

$\Rightarrow x \ln x = ye^y$

$\Rightarrow \alpha \ln \alpha = 2e^2$

$\alpha^\alpha = e^{2e^2}$

15. The slope of tangent at any point (x, y) on a curve $y = y(x)$ is $\frac{x^2+y^2}{2xy}$, $x > 0$. If $y(2) = 0$, then a value of $y(8)$ is:

[JEE(Main)April,2023]

वक्र $y = y(x)$ के किसी भी बिंदु (x, y) पर स्पर्श रेखा की प्रवणता $\frac{x^2+y^2}{2xy}$, $x > 0$ है। यदि $y(2) = 0$ है, तो $y(8)$ का एक मान है :

[JEE(Main)April,2023]

(A) $-4\sqrt{2}$ (B) $4\sqrt{3}$ (C) $2\sqrt{3}$ (D) $-2\sqrt{3}$

Ans. B

Sol. $\frac{dy}{dx} = \frac{x^2+y^2}{2xy}$

Let $y = \upsilon x \therefore \frac{dy}{dx} = \upsilon + x\frac{d\upsilon}{dx}$ $\therefore \upsilon + x\frac{d\upsilon}{dx} = \frac{x^2+\upsilon^2x^2}{2x.\upsilon x}$

$x\frac{d\upsilon}{dx} = \frac{1+\upsilon^2}{2\upsilon} - \upsilon = \frac{1-\upsilon^2}{2\upsilon} \Rightarrow \frac{2\upsilon}{1-\upsilon^2}d\upsilon = \frac{dx}{x} \Rightarrow \int\frac{2\upsilon d\upsilon}{\upsilon^2-1} + \int\frac{dx}{x} = 0$

$\Rightarrow \ell n\left(\upsilon^2-1\right) + \ell nx = \ell nc \Rightarrow x\left(\upsilon^2-1\right) = c \Rightarrow x\frac{y^2-x^2}{x^2} = c \Rightarrow y^2 - x^2 = cx$

Now $f(2) = 0 \therefore c = -2$

$\therefore y^2 - x^2 = -2x$ $\therefore x = 8$ then $y^2 - 64 = -16$

$y^2 = 48$

$y = f(8) = 4\sqrt{3}$

16. Let $y = y(x)$ be a solution curve of the differential equation $(1 - x^2y^2)dx = ydx + xdy$. If the line $x = 1$ intersects the curve $y = y(x)$ at $y = 2$ and the line $x = 2$ intersects the curve $y = y(x)$ at $y = \alpha$, then a value of α is :

[JEE(Main)April,2023]

माना अवकल समीकरण $(1 - x^2y^2)dx = ydx + xdy$ का हल वक्र $y = y(x)$ है। यदि रेखा $x = 1$, वक्र $y = y(x)$ को $y = 2$ पर काटती gS rFkk ljy js[kk $x = 2$, वक्र $y = y(x)$ को $y = \alpha$ पर काटती है, तो α का एक मान है : **[JEE(Main)April,2023]**

(A) $\frac{3e^2}{2\left(3e^2+1\right)}$ (B) $\frac{3e^2}{2\left(3e^2-1\right)}$ (C) $\frac{1+3e^2}{2\left(3e^2-1\right)}$ (D) $\frac{1-3e^2}{2\left(3e^2+1\right)}$

Ans. C

Sol. $dx = \frac{d(xy)}{1-(xy)^2} \Rightarrow 2dx = \frac{d(xy)}{1-xy} + \frac{d(xy)}{1+xy}$

$$2x + c = \ell n\left|\frac{1+xy}{1-xy}\right| \Rightarrow \left|\frac{xy+1}{xy-1}\right| = e^{c}\cdot e^{2x}$$

$\because y(1) = 2$

$\Rightarrow 3 = e^{c}\cdot e^{2}$

$\Rightarrow e^{c} = \frac{3}{e^{2}}e^{2x}$

So at $x = 2$

$\Rightarrow \left|\frac{2\alpha+1}{2\alpha-1}\right| = 3e^{2}$

Case : 1 for +ve sign

$\Rightarrow \frac{2\alpha+1}{2\alpha-1} = \frac{3e^{2}}{1}$

$\Rightarrow \frac{4\alpha}{2} = \frac{3e^{2}+1}{3e^{2}-1}$

$\Rightarrow \alpha = \frac{1}{2}\left(\frac{3e^{2}+1}{3e^{2}-1}\right)$

Case: 2 for (–ve)sign not required

17. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{5}{x\left(x^{5}+1\right)}y = \frac{\left(x^{5}+1\right)^{2}}{x^{7}}, x > 0$. If $y(1) = 2$, then $y(2)$ is equal to : **[JEE(Main)April,2023]**

माना अवकल समीकरण $\frac{dy}{dx} + \frac{5}{x\left(x^{5}+1\right)}y = \frac{\left(x^{5}+1\right)^{2}}{x^{7}}, x > 0$ का हल $y = y(x)$ है। यदि $y(1) = 2$ है, तो $y(2)$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{679}{128}$ (B) $\frac{637}{128}$ (C) $\frac{697}{128}$ (D) $\frac{693}{128}$

Ans. D

Sol. $I.F = e^{\int\frac{5dx}{x\left(x^{5}+1\right)}} = e^{\frac{5x^{-6}}{\left(x^{-5}+1\right)}}$

$Put, 1 + x^{-5} = t \Rightarrow -5x^{-6}dx = dt$

$\Rightarrow e^{\int\frac{-dt}{t}} = \frac{1}{t} = \frac{x^{5}}{1+x^{5}}$

$y\cdot\frac{x^{5}}{1+x^{5}} = \int\frac{x^{5}}{\left(1+x^{5}\right)}\times\frac{\left(1+x^{5}\right)^{2}}{x^{7}}dx$

$$= \int x^3 dx + \int x^{-2} dx$$

$$y \cdot \frac{x^5}{1+x^5} = \frac{x^4}{4} - \frac{1}{x} + c$$

Given than : $x = 1 \Rightarrow y = 2$

$$2 \cdot \frac{1}{2} = \frac{1}{4} - 1 + c$$

$$c = \frac{7}{4}$$

$$y \cdot \frac{x^5}{1+x^5} = \frac{x^4}{4} - \frac{1}{x} + \frac{7}{4}$$

Now put, $x = 2$

$$y \cdot \left(\frac{32}{33}\right) = \frac{21}{4}$$

$$y = \frac{693}{128}$$

18. Let $y = y(x)$, $y > 0$, be a solution curve of the differential equation $\left(1+x^2\right)dy = y(x-y)dx$. If $y(0) = 1$ and $y(2\sqrt{2}) = \beta$, then : **[JEE(Main)April,2023]**

माना अवकल समीकरण $\left(1+x^2\right)dy = y(x-y)dx$ का एक हल वक्र $y = y(x)$, $y > 0$ है। यदि $y(0) = 1$ तथा $y(2\sqrt{2}) = \beta$ हैं, r ks **[JEE(Main)April,2023]**

(A) $e^{\beta^{-1}} = e^{-2}(3+2\sqrt{2})$ (B) $e^{3\beta^{-1}} = e(3+2\sqrt{2})$

(C) $e^{3\beta^{-1}} = e(5+\sqrt{2})$ (D) $e^{\beta^{-1}} = e^{-2}(5+\sqrt{2})$

Ans. B

Sol. $y(x-y)dx = \left(1+x^2\right)dy$

$$\frac{y(x-y)}{1+x^2} = \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{x}{1+x^2}y - \frac{y^2}{1+x^2}$$

$$\frac{dy}{dx} - \frac{x}{1+x^2}y = \frac{-y^2}{1+x^2}$$

$$\frac{-1}{y^2}\frac{dy}{dx} + \frac{x}{1+x^2}\frac{1}{y} = \frac{1}{1+x^2}$$

Let $\frac{1}{y} = t$

$$\therefore \frac{-1}{y^2}\frac{dy}{dx} = \frac{dt}{dx}$$

$$\therefore \frac{dt}{dx} + \frac{x}{1+x^2}t = \frac{1}{1+x^2} \text{ linear in t}$$

I.F. $= e^{\int \frac{x}{1+x^2}dx} = e^{\frac{1}{2}\ln\left(1+x^2\right)} = \sqrt{1+x^2}$

$\therefore$ solution is

$$t\sqrt{1+x^2} = \int \frac{\sqrt{1+x^2}}{1+x^2}dx + c$$

$$\frac{1}{y}\sqrt{1+x^2} = \int \frac{1}{\sqrt{1+x^2}}dx + c$$

$$\frac{1}{y}\sqrt{1+x^2} = \ln\left(x + \sqrt{x^2+1}\right) + c$$

$$y(0) = 1 \quad \Rightarrow \quad 1 = 0 + c \quad \Rightarrow \quad c = 1$$

So at $x = 2\sqrt{2}$ we have $\frac{3}{\beta} = \ln(2\sqrt{2}+3)+1 \Rightarrow 3\beta^{-1} = \ln e(2\sqrt{2}+3)$

$$\Rightarrow e^{3\beta^{-1}} = e(3+2\sqrt{2})$$

19. Let $y = y_1(x)$ and $y = y_2(x)$ be the solution curves of the differential equation $\frac{dy}{dx} = y + 7$ with initial conditions $y_1(0) = 0$ and $y_2(0) = 1$ respectively. Then the curves $y = y_1(x)$ and $y = y_2(x)$ intersect at :

[JEE(Main)April,2023]

(A) two points (B) infinite number of points

(C) no point (D) one point

माना अवकल समीकरण $\frac{dy}{dx} = y + 7$ के प्रारंभिक प्रतिबंधो $y_1(0) = 0$ तथा $y_2(0) = 1$ के साथ हल वक्र क्रमशः $y = y_1(x)$ तथा $y = y_2(x)$ हैं। तो वक्र $y = y_1(x)$ तथा $y = y_2(x)$: **[JEE(Main)April,2023]**

(A) दो बिन्दुओं पर मिलते हैं (B) अनंत बिन्दुओं पर मिलते हैं

(C) किसी भी बिन्दु पर नहीं मिलते (D) एक बिन्दु पर मिलते हैं

Ans. C

Sol. $\frac{dy}{dx} = y + 7 \Rightarrow \frac{dy}{dx} - y = 7$

I.F. $= e^{-x}$

$$ye^{-x} = \int 7e^{-x}dx$$

$$\Rightarrow ye^{-x} = -7e^{-x} + c$$

$$\Rightarrow y = -7 + ce^{x}$$

$-7+7e^x=-7+8e^x \Rightarrow e^x=0$

No solution.

20. Let x = x(y) be the solution of the differential equation

$2(y+2)\log_e(y+2)dx+\left(x+4-2\log_e(y+2)\right)dy=0, y>-1$ with $x\left(e^4-2\right)=1$. Then $x\left(e^9-2\right)$ is equal to :

[JEE(Main)April,2023]

यदि x = x(y) अवकल समीकरण $2(y+2)\log_e(y+2)dx+\left(x+4-2\log_e(y+2)\right)dy=0, y>-1$ का हल है जहाँ $x\left(e^4-2\right)=1$ हो, तो $x\left(e^9-2\right)$ का मान बराबर होगा :

[JEE(Main)April,2023]

(A) $\frac{10}{3}$ (B) $\frac{4}{9}$ (C) 3 (D) $\frac{32}{9}$

Ans. D

Sol. $2(y+2)\ln(y+2)dx+(x+4-2\ln(y+2))dy=0$

$$2\ln\left(y+2\right)+\left(x+4-2\ln\left(y+2\right)\right)\frac{1}{y+2}\cdot\frac{dy}{dx}=0$$

let $\ln(y+2)=t$

$$\frac{1}{y+2}\cdot\frac{dy}{dx}=\frac{dt}{dx}$$

$$2t+\left(x+4-2t\right)\cdot\frac{dt}{dx}=0$$

$$\left(x+4-2t\right)\frac{dt}{dx}=-2t$$

$$\frac{dx}{dt}=\frac{2t-4-x}{2t}$$

$$\frac{dx}{dt}+\frac{x}{2t}=\frac{2t-4}{2t}$$

$$x\cdot t^{1/2}=\int\frac{2t-4}{2t}\cdot t^{1/2}\cdot dt$$

$$x\cdot t^{1/2}=\int\left(t^{1/2}-\frac{2}{t^{1/2}}\right)\cdot dt$$

$$xt^{\frac{1}{2}}=\frac{t^{\frac{3}{2}}}{\frac{3}{2}}-2\cdot\frac{t^{\frac{1}{2}}}{\frac{1}{2}}+C$$

$$x\cdot t^{\frac{1}{2}}=\frac{2t^{\frac{3}{2}}}{3}-4t^{\frac{1}{2}}+C$$

$$x = \frac{2}{3}\cdot t - 4 + C\cdot t^{-\frac{1}{2}}$$

$$x = \frac{2}{3}\ln(y+2) - 4 + C\cdot(\ln(y+2))^{-\frac{1}{2}}$$

Put $y = e^4 - 2$, $x = 1$

$$1 = \frac{2}{3}\times 4 - 4 + C\times\frac{1}{2}$$

$$\frac{C}{2} = 5 - \frac{8}{3} = \frac{7}{3}$$

$$\Rightarrow C = \frac{14}{3}$$

$$x = \frac{2}{3}\times 9 - 4 + \frac{14}{3}\times\frac{1}{3}$$

$$= 2 + \frac{14}{9}$$

$$= \frac{32}{9}$$

21. Let $y = y(x)$ be a solution of the differential equation $(x\cos x)dy + (xy\sin x + y\cos x - 1)dx = 0, 0 < x < \frac{\pi}{2}$. If $\frac{\pi}{3}y\left(\frac{\pi}{3}\right) = \sqrt{3}$, then $\left|\frac{\pi}{6}y''\left(\frac{\pi}{6}\right) + 2y'\left(\frac{\pi}{6}\right)\right|$ is equal to _____. **[JEE(Main)April,2023]**

माना अवकल समीकरण $(x\cos x)dy + (xy\sin x + y\cos x - 1)dx = 0, 0 < x < \frac{\pi}{2}$ का हल $y = y(x)$ है। यदि $\frac{\pi}{3}y\left(\frac{\pi}{3}\right) = \sqrt{3}$ है, तब $\left|\frac{\pi}{6}y''\left(\frac{\pi}{6}\right) + 2y'\left(\frac{\pi}{6}\right)\right|$ का मान है _____ **[JEE(Main)April,2023]**

Ans. B

Sol. $(x\cos x)dy + (xy\sin x + y\cos x - 1)dx = 0, 0 < x < \frac{\pi}{2}$

$$\frac{dy}{dx} + \left(\frac{x\sin x + \cos x}{x\cos x}\right)y = \frac{1}{x\cos x}$$

IF = x secx

$$y\cdot x\sec x = \int\frac{x\sec x}{x\cos x}dx = \tan x + c$$

Since $y\left(\frac{\pi}{3}\right) = \frac{3\sqrt{3}}{\pi}$ Hence $c = \sqrt{3}$

Hence $\left|\frac{\pi}{6}y''\left(\frac{\pi}{6}\right)+y'\left(\frac{\pi}{6}\right)\right|=|-2|=2$

22. Let a curve $y=f(x), x\in(0,\infty)$ pass through the points $P\left(1,\frac{3}{2}\right)$ and $Q\left(a,\frac{1}{2}\right)$. If the tangent at any point R(b, f(b)) to the given curve cuts the y-axis at the point S(0, c) such that bc = 3, then $(PQ)^2$ is equal to _____.

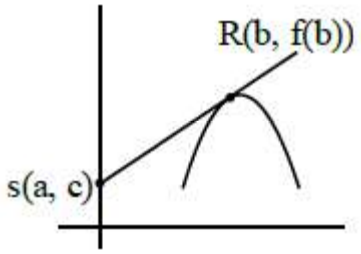
[JEE(Main)April,2023]

माना एक वक्र $y=f(x), x\in(0,\infty)$ बिंदुओं $P\left(1,\frac{3}{2}\right)$ तथा $Q\left(a,\frac{1}{2}\right)$ से होकर जाता है। यदि दिए गए वक्र के किसी भी बिंदु R(b, f(b)) पर स्पर्श रेखा y-अक्ष को बिंदु S(0, c) पर काटती है जबकि bc = 3 तो $(PQ)^2$ बराबर है _____

[JEE(Main)April,2023]

Ans. 5

Sol.



Equation of tangent at R(b, f(2)) is

$y-f(b)=f'(b).(x-b)$

which passes through (0, c)

$\Rightarrow c-f(b)=f'(b)\cdot(-b)$

$\Rightarrow \frac{3}{b}-f(b)=f'(b)\cdot(-b)$

$\Rightarrow bf'(b)-f(b)=-\frac{3}{b}$

$\Rightarrow \frac{bf'(b)-f(b)}{b^2}=-\frac{3}{b^3}$

$\Rightarrow d\left(\frac{f(b)}{b}\right)=-\frac{3}{b^3}\Rightarrow \frac{f(b)}{b}=\frac{3}{2b^2}+\lambda$

Which passes through (1, 3/2)

$\Rightarrow \frac{3}{2}=\frac{3}{2}+\lambda\Rightarrow\lambda=0$

$\Rightarrow f(b)=\frac{3}{2b}$

$f(a)=\frac{1}{2}\Rightarrow\frac{1}{2}=\frac{3}{2b}\Rightarrow b=3$

$\Rightarrow c = 1 \Rightarrow Q(3, 1/2)$

$\Rightarrow PQ^2 = 2^2 + (1)^2 = 5$

23. If the solution curve of the differential equation $\left(y - 2\log_e x\right)dx + \left(x \log_e x^2\right)dy = 0, x > 1$ passes through the points $\left(e, \frac{4}{3}\right)$ and (e^4, α), then α is equal to _____. **[JEE(Main)April,2023]**

यदि अवकल समीकरण $\left(y - 2\log_e x\right)dx + \left(x \log_e x^2\right)dy = 0, x > 1$ का हल वक्र बिंदुओं $\left(e, \frac{4}{3}\right)$ तथा (e^4, α) से होकर t kr kgrk gS] rks α बराबर है _____ **[JEE(Main)April,2023]**

Ans. 3

Sol. $\because \quad (y - 2\ln x)dx + (2x \ln x)dy = 0.$

$$2x \ln x \frac{dy}{dx} + y = 2\ln x$$

$$\frac{dy}{dx} + \frac{y}{2x \ln x} = \frac{1}{x}$$

$$\therefore \quad \text{I.F.} = e^{\int \frac{1}{2x \ln x} dx} = \sqrt{\ln x}$$

$\therefore$ Solution of the equation is :

$$y \cdot \sqrt{\ln x} = \int \frac{\sqrt{\ln x}}{x} dx$$

$$\therefore \quad y \cdot \sqrt{\ln x} = \frac{2}{3}(\ln x)^{\frac{3}{2}} + C \quad \ldots \text{(i)}$$

$\therefore$ eq. (i) passes through point $\left(e, \frac{4}{3}\right)$.

$$\therefore C = \frac{2}{3}$$

$$\therefore y\sqrt{\ln x} = \frac{2}{3}(\ln x)^{\frac{3}{2}} + \frac{2}{3} \quad \ldots \text{(ii)}$$

This equation passes through point (e^4, α)

$\therefore \alpha = 3.$

24. Let the solution curve $x = x(y), 0 < y < \frac{\pi}{2}$, of the differential equation

$\left(\log_e(\cos y)\right)^2 \cos y\, dx - \left(1 + 3x \log_e(\cos y)\right)\sin y dy = 0$ satisfy $x\left(\frac{\pi}{3}\right) = \frac{1}{2\log_e 2}$. If

$x\left(\frac{\pi}{6}\right) = \frac{1}{\log_e m - \log_e n}$, where m and n are coprime, then mn is equal to _____.

माना अवकल समीकरण $\left(\log_e(\cos y)\right)^2 \cos y\,dx - \left(1+3x\log_e(\cos y)\right)\sin y dy = 0$, का हल वक्र x = x(y), $0 < y < \frac{\pi}{2}$;

$x\left(\frac{\pi}{3}\right) = \frac{1}{2\log_e 2}$ को संतुष्ट करता है। यदि $x\left(\frac{\pi}{6}\right) = \frac{1}{\log_e m - \log_e n}$ है, जहाँ m तथा n असहभाज्य है, तो mn बराबर है :

[JEE(Main)April,2023]

Ans. 12

Sol. $\cos y \ln^2 \cos y dx = (1+3x\ln\cos y)\sin y dy$

$$\frac{dx}{dy} = \tan y\left(\frac{3x}{\ln\cos y} + \frac{1}{\ln^2\cos y}\right)$$

$$\frac{dx}{dy} - \left(\frac{3\tan y}{\ln\cos y}\right)x = \frac{\tan y}{\ln^2\cos y}$$

$$IF = e^{\int\frac{-3\sin y}{\ln\cos y\cos y}dy}$$

$$\ln\cos y = t$$

$$\frac{1}{\cos y}(-\sin y)dy = dt$$

$$If = e^{\int\frac{3}{t}dt} = e^{3\ln t} = t^3 = \ln^3\cos y$$

$$\text{solution is } x\cdot\ln^3\cos y = \frac{\sin y}{\cos y}\cdot\ln\cos y dy + C$$

$$x\ln^3\cos y = \frac{-\ln^2\cos y}{2} + C$$

$$x\left(\frac{\pi}{3}\right) = \frac{1}{2\ln 2} \text{ so } \frac{1}{2\ln 2}\times\ln^3\left(\frac{1}{2}\right) = -\frac{\ln^2\left(\frac{1}{2}\right)}{2} + C$$

$$C = 0$$

$$y = \frac{\pi}{6} \Rightarrow x\ln^3\frac{\sqrt{3}}{2} = -\frac{1}{2}\ln^2\frac{\sqrt{3}}{2} + 0$$

$$x = -\frac{1}{2\ln\left(\frac{\sqrt{3}}{2}\right)}$$

$$x = \frac{1}{\ln\frac{4}{3}} = \frac{1}{\ln 4 - \ln 3}$$

mn = 12

25. Let the tangent at any point P on a curve passing through the points (1, 1) and $\left(\frac{1}{10},100\right)$, intersect positive x-axis and y-axis at the points A and B respectively. If PA : PB = 1: k and y = y(x) is the solution of the differential equation $e^{\frac{dy}{dx}} = kx + \frac{k}{2}, y(0) = k$, then $4y(1) - 5\log_e 3$ is equal to _____. **[JEE(Main)April,2023]**

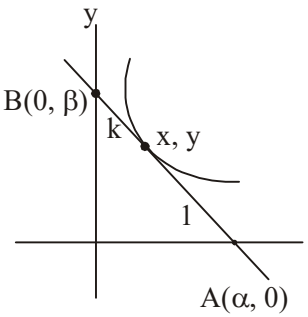
माना बिंदुओं (1, 1) तथा $\left(\frac{1}{10},100\right)$ से होकर जाने वाले एक वक्र के किसी बिंदु P पर स्पर्श रेखा धनात्मक x-अक्ष तथा y-अक्षों को Øe' l?fcaqlA तथा B पर काटती है। यदि PA : PB = 1: k है तथा अवकल समीकरण $e^{\frac{dy}{dx}} = kx + \frac{k}{2}, y(0) = k$ का हल y = y(x) है, तो $4y(1) - 5\log_e 3$ बराबर है : **[JEE(Main)April,2023]**

Ans. Bonus

Sol. $Y - y = \frac{dy}{dx}(X - x)$

$Y = 0$

$X = \frac{-ydx}{dy} + x$



$\frac{k\alpha + 0}{k+1} = 0, \quad \alpha = \frac{k+1}{k}x$

$\frac{k+1}{k}x = -y\frac{dx}{dy} + x$

$x + \frac{x}{k} = -y\frac{dx}{dy} + x$

$x\frac{dy}{dx} + ky = 0 \quad \frac{dy}{dx} + \frac{k}{x}y = 0$

$y.x^k = C$

$C = 1$

$$100\cdot\left(\frac{1}{10}\right)^{k}=1$$

$K=2$

$$\frac{dy}{dx}=\ln(2x+1)$$

$$y=\frac{(2x+1)}{2}(\ln(2x+1)-1)+c$$

$$2=\frac{1}{2}(0-1)+C$$

$$C=2+\frac{1}{2}=\frac{5}{2}$$

$$y(1)=\frac{3}{2}(\ln 3-1)+\frac{5}{2}$$

$$=\frac{3}{2}\ln 3+1$$

$$4y(1)=6\ln 3+4$$

$$4y(1)-5\ln 3=4+\ell\ln 3$$

26. If y = y(x) is the solution of the differential equation $\frac{dy}{dx}+\frac{4x}{\left(x^2-1\right)}y=\frac{x+2}{\left(x^2-1\right)^{\frac{5}{2}}},x>1$ such that $y(2)=\frac{2}{9}\log_e(2+\sqrt{3})$ and $y(\sqrt{2})=\alpha\log_e(\sqrt{\alpha}+\beta)+\beta-\sqrt{\gamma},\alpha,\beta,\gamma\in\mathbb{N}$, then $\alpha\beta\gamma$ is equal to _____.

[JEE(Main)April,2023]

यदि $\frac{dy}{dx}+\frac{4x}{\left(x^2-1\right)}y=\frac{x+2}{\left(x^2-1\right)^{\frac{5}{2}}},x>1$ का हल y = y(x) है तथा $y(2)=\frac{2}{9}\log_e(2+\sqrt{3})$ और $y(\sqrt{2})=\alpha\log_e(\sqrt{\alpha}+\beta)+\beta-\sqrt{\gamma},\alpha,\beta,\gamma\in\mathbb{N}$ हैं, तो $\alpha\beta\gamma$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 6

Sol. $\frac{dy}{dx}+\frac{4x}{\left(x^2-1\right)}y=\frac{x+2}{\left(x^2-1\right)^{\frac{5}{2}}},x>1$ which is linear in y

I.f. $e^{\int\frac{4x}{x^2-1}dx}=e^{2\ln\left(x^2-1\right)}=\left(x^2-1\right)^2$

Solution is $y\cdot\left(x^2-1\right)^2=\int\frac{x+2}{\left(x^2-1\right)^{\frac{5}{2}}}\left(x^2-1\right)^2dx+c$

$$\Rightarrow y\left(x^2-1\right)^2=\int\frac{x+2}{\sqrt{x^2-1}}dx+c$$

$$\Rightarrow y\left(x^2-1\right)^2=\sqrt{x^2-1}+2\ln\left(x+\sqrt{x^2-1}\right)+c$$

$$\therefore y(2)=\frac{2}{9}\ln\left(2+\sqrt{3}\right)$$

$$\therefore \frac{2}{9}\ln\left(2+\sqrt{3}\right)\cdot 9=\sqrt{3}+2\ln\left(2+\sqrt{3}\right)+c$$

$$c=-\sqrt{3}$$

$$\therefore y\left(x^2-1\right)^2=\sqrt{x^2-1}+2\ln\left(x+\sqrt{x^2-1}\right)-\sqrt{3}$$

Put $x=\sqrt{2}$

$$\therefore y\left(\sqrt{2}\right)\cdot 1=1+2\ln\left(\sqrt{2}+1\right)-\sqrt{3}$$

$$y\left(\sqrt{2}\right)=2\ln\left(\sqrt{2}+1\right)+1-\sqrt{3}$$

$$=\alpha\ln\left(\sqrt{\alpha}+\beta\right)+\beta-\sqrt{\gamma}$$

$\alpha = 2, \beta = 1, \gamma = 3$ $\quad\therefore \alpha\cdot\beta\cdot\gamma = 6$

27. Let $x = x(t)$ and $y = y(t)$ be solutions of the differential equations $\frac{dx}{dt}+ax=0$ and $\frac{dy}{dt}+by=0$ respectively, a, b $\in$ R. Given that $x(0) = 2$; $y(0) = 1$ and $3y(1) = 2x(1)$, the value of t, for which $x(t) = y(t)$, is :
[JEE(Main)Jan.2024]

माना अवकल समीकरणों $\frac{dx}{dt}+ax=0$ तथा $\frac{dy}{dt}+by=0$, a, b $\in$ R के हल क्रमशः x = x(t) तथा y = y(t) हैं। यदि $x(0) = 2$; $y(0) = 1$ तथा $3y(1) = 2x(1)$ हैं, तो $x(t) = y(t)$ के लिए t का मान है : **[JEE(Main)Jan.2024]**

(A) $\log_3 4$ (B) $\log_{\frac{2}{3}} 2$ (C) $\log_4 3$ (D) $\log_{\frac{4}{3}} 2$

Ans. D

Sol. $\frac{dx}{dt}+ax=0$

$$\Rightarrow \int\frac{1}{x}dx=-\int a\,dt$$

$\Rightarrow \ln|x| = -at + c$

$\Rightarrow x = \pm e^c \cdot e^{-at} = \lambda e^{-at}$

$\Rightarrow x(0) = 2 \quad \Rightarrow \lambda = 2$

$\Rightarrow x = 2e^{-at} \quad$ _____(1)

Now, $\frac{dy}{dt}+by=0$

$\Rightarrow \int \frac{1}{y} dy = -\int b\, dt \quad \Rightarrow ln(y) = -bt + c$

$\Rightarrow y = \pm e^c \cdot e^{-bt} = \mu \cdot e^{-bt}$

$y(0) = 1$

$\therefore \mu = e^{-bt}$

$3y(1) = 2x(1)$

$3e^{-b} = 2(2e^{-a}) = 4e^{-a}$

$e^{-b+a} = \frac{4}{3}$

$\Rightarrow x(t) = y(t)$

$\Rightarrow e^{-bt} = 2e^{-at}$

$\Rightarrow e^{-bt+at} = 2$

$\Rightarrow e^{(-b+a)t} = 2$

$\Rightarrow \left(\frac{4}{3}\right)^t = 2$

$\Rightarrow t\ell n\frac{4}{3} = \ell n2$

$\Rightarrow t = \log_{\frac{4}{3}} 2$

28. If $y = y(x)$ is the solution curve of the differential equation $\left(x^2-4\right)dy - \left(y^2-3y\right)dx = 0, x > 2, y(4) = \frac{3}{2}$ and the slope of the curve is never zero, then the value of $y(10)$ equals : **[JEE(Main)Jan.2024]**

यदि अवकल समीकरण $\left(x^2-4\right)dy - \left(y^2-3y\right)dx = 0, x > 2, y(4) = \frac{3}{2}$ का हल वक्र $y = y(x)$ है तथा इस वक्र की प्रवणता fd l hHhx के लिए शून्य नहीं है, तो $y(10)$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) $\frac{3}{1+2\sqrt{2}}$ (B) $\frac{3}{1+(8)^{1/4}}$ (C) $\frac{3}{1-2\sqrt{2}}$ (D) $\frac{3}{1-(8)^{1/4}}$

Ans. B

Sol. $(x^2-4)dy - (y^2-3y)dx = 0$

$\Rightarrow \int \frac{dy}{y^2-3y} = \int \frac{dx}{x^2-4}$

$\Rightarrow \frac{1}{3}\int \frac{y-(y-3)}{y(y-3)}dy = \int \frac{dx}{x^2-4}$

$\Rightarrow \frac{1}{3}(\ell n|y-3| - \ell n|y|) = \frac{1}{4}\ell n\left|\frac{x-2}{x+2}\right| + C$

$$\Rightarrow \frac{1}{3}\ell n\left|\frac{y-3}{y}\right| = \frac{1}{4}\ell n\left|\frac{x-2}{x+2}\right| + C$$

At $x = 4,\ y = \frac{3}{2}$

$$\therefore C = \frac{1}{4}\ell n3$$

$$\therefore \frac{1}{3}\ell n\left|\frac{y-3}{y}\right| = \frac{1}{4}\ell n\left|\frac{x-2}{x+2}\right| + \frac{1}{4}\ell n(3)$$

At $x = 10$

$$\frac{1}{3}\ell n\left|\frac{y-3}{y}\right| = \frac{1}{4}\ell n\left|\frac{2}{3}\right| + \frac{1}{4}\ell n(3)$$

$$\ell n\left|\frac{y-3}{y}\right| = \ell n2^{3/4}, \forall x > 2, \frac{dy}{dx} < 0$$

as $y(4) = \frac{3}{2} \Rightarrow y \in (0,3)$

$$-y + 3 = 8^{1/4} \cdot y$$

$$y = \frac{3}{1+8^{1/4}}$$

29. A function $y = f(x)$ satisfies $f(x)\sin 2x + \sin x - (1+\cos^2 x)f'(x) = 0$ with condition $f(0) = 0$. Then, $f\left(\frac{\pi}{2}\right)$ is equal to : **[JEE(Main)Jan.2024]**

एक फलन $y = f(x)$ के लिए $f(x)\sin 2x + \sin x - (1+\cos^2 x)f'(x) = 0$ तथा $f(0) = 0$ हैं। तो $f\left(\frac{\pi}{2}\right)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 2 (B) 1 (C) 0 (D) – 1

Ans. B

Sol. $$\frac{dy}{dx} - \left(\frac{\sin 2x}{1+\cos^2 x}\right)y = \frac{\sin x}{1+\cos^2 x}$$

$$\text{If} = e^{-\int \frac{\sin 2x}{1+\cos^2 x}dx} = e^{\ln\left(1+\cos^2 x\right)} = 1+\cos^2 x$$

So general solution of DE

$$y\left(1+\cos^2 x\right)=\int \frac{\sin x}{1+\cos^2 x}\cdot\left(1+\cos^2 x\right)dx$$

$$y\left(1+\cos^2 x\right)=-\cos x+c$$

$$y\left(1+\cos^2 x\right)+\cos x=c$$

were $y(0)=0$

$0+1=c$

$$y\left(1+\cos^2 x\right)+\cos x=1$$

$$y=\frac{1-\cos x}{1+\cos^2 x}$$

$$y\left(\frac{\pi}{2}\right)=\frac{1-0}{1+0}=1$$

30. If $\sin\left(\frac{y}{x}\right)=\log_e|x|+\frac{\alpha}{2}$ is the solution of the differential equation $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx}=y\cos\left(\frac{y}{x}\right)+x$ and $y(1)=\frac{\pi}{3}$, then α^2 is equal to : **[JEE(Main)Jan.2024]**

यदि अवकल समीकरण $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx}=y\cos\left(\frac{y}{x}\right)+x$, $y(1)=\frac{\pi}{3}$, का हल $\sin\left(\frac{y}{x}\right)=\log_e|x|+\frac{\alpha}{2}$ है, तो α^2 बराबर है : **[JEE(Main)Jan.2024]**

(A) 3 (B) 9 (C) 12 (D) 4

Ans. A

Sol. Given differential equation is $x\cos\frac{y}{x}\frac{dy}{dx}-y\cos\frac{y}{x}=x$

$$\Rightarrow \cos\frac{y}{x}\left(x\frac{dy}{dx}-y\right)=x$$

$$\Rightarrow \cos\frac{y}{x}\left(\frac{x\frac{dy}{dx}-y\cdot 1}{x^2}\right)=\frac{1}{x}$$

Let $\frac{y}{x}=t$

$$\cos t\frac{dt}{dx}=\frac{1}{x}$$

$$\cos t\,dt=\frac{1}{x}dx$$

on integeration $\sin t = \ell n|x| + c$

$y(1) = \frac{\pi}{3} \Rightarrow \sin\frac{\pi}{3} = 0 + c$

$\Rightarrow c = \frac{\sqrt{3}}{2}$

$\therefore$ solution is $\sin\frac{y}{x} = \ell n|x| + \frac{\sqrt{3}}{2}$

31. Let y = y(x) be the solution of the differential equation $\sec x dy + \{2(1-x)\tan x + x(2-x)\}dx = 0$ such that $y(0) = 2$. Then $y(2)$ is equal to : **[JEE(Main)Jan.2024]**

माना अवकल समीकरण $\sec x dy + \{2(1-x)\tan x + x(2-x)\}dx = 0$, $y(0) = 2$ का हल $y = y(x)$ है। तो $y(2)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 1 (B) $2\{\sin(2) + 1\}$ (C) 2 (D) $2\{1 - \sin(2)\}$

Ans. C

Sol. $dy = \frac{2(x-1)\tan x + x(x-2)}{\sec x}dx$

$\Rightarrow \int dy = \int 2(x-1)\sin x + (x^2 - 2x)\cos x$

$\Rightarrow y = 2[(x-1)(-\cos x) - (1)(-\sin x)] + [(x^2-2x)(\sin x) - (2x-2)(-\cos x) + (2)(-\sin x)] + c$

$y(0) = 2[1] + [-2] + C$

$y(0) = 2$

$C = 2$

$y(2) = 2(1 + \sin 2) - 2\sin 2 = 2$

32. Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} = \frac{(\tan x) + y}{\sin x(\sec x - \sin x \tan x)}$, $x \in \left(0, \frac{\pi}{2}\right)$ satisfying the condition $y\left(\frac{\pi}{4}\right) = 2$. Then $y\left(\frac{\pi}{3}\right)$, is : **[JEE(Main)Jan.2024]**

माना अवकल समीकरण $\frac{dy}{dx} = \frac{(\tan x) + y}{\sin x(\sec x - \sin x \tan x)}$, $x \in \left(0, \frac{\pi}{2}\right)$ के हल $y = y(x)$ के लिए $y\left(\frac{\pi}{4}\right) = 2$ है। तो $y\left(\frac{\pi}{3}\right)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) $\sqrt{3}\left(2 + \log_e\sqrt{3}\right)$ (B) $\frac{\sqrt{3}}{2}\left(2 + \log_e 3\right)$ (C) $\sqrt{3}\left(2 + \log_e 3\right)$ (D) $\sqrt{3}\left(1 + 2\log_e 3\right)$

Ans. A

Sol. $\dfrac{dy}{dx}=\dfrac{\sin x+y\cos x}{\sin x\cdot\cos x\left(\dfrac{1}{\cos x}-\sin x\dfrac{\sin x}{\cos x}\right)}$

$=\dfrac{\sin x+y\cos x}{\sin x\left(1-\sin^2 x\right)}$

$\dfrac{dy}{dx}=\sec^2 x+y\cdot 2\left(\text{cosec}2x\right)$

$\dfrac{dy}{dx}-2\text{cosec}\left(2x\right)\cdot y=\sec^2 x$

$\dfrac{dy}{dx}+p\cdot y=Q$

$\text{I.F.}=e^{\int p dx}=e^{\int -2\,\text{cosec}(2x)dx}$

Let $2x=t$

$2\dfrac{dx}{dt}=1$

$dx=\dfrac{dt}{2}$

$=e^{-\int \text{cosec}(t)dt}$

$=e^{-\ell n\left|\tan\frac{t}{2}\right|}$

$=e^{-\ell n|\tan x|}=\dfrac{1}{|\tan x|}$

$y(\text{IF})=\int Q(\text{IF})dx+c$

$\Rightarrow y\dfrac{1}{|\tan x|}=\int \sec^2 x\cdot\dfrac{1}{|\tan x|}+c$

$y\cdot\dfrac{1}{|\tan x|}=\int \dfrac{dt}{|t|}+c$ for $\tan x=t$

$y\cdot\dfrac{1}{|\tan x|}=\ell n|t|+c$

$y=|\tan x|\left(\ell n|\tan x|+c\right)$

Put $x=\dfrac{\pi}{4},\ y=2$

$2=\ell n\,1+c\Rightarrow c=2$

$y=|\tan x|\,(\ell n\,|\tan x|+2)$

$$y\left(\frac{\pi}{3}\right)=\sqrt{3}\left(\ell n\sqrt{3}+2\right)$$

33. The solution curve of the differential equation $y\frac{dx}{dy}=x\left(\log_e x-\log_e y+1\right)$, $x>0$, $y>0$ passing through the point (e, 1) is : **[JEE(Main)Jan.2024]**

बिन्दु (e, 1) से होकर जाने वाला, अवकल समीकरण $y\frac{dx}{dy}=x\left(\log_e x-\log_e y+1\right)$, $x>0$, $y>0$ का हल वक्र है : **[JEE(Main)Jan.2024]**

(A) $\left|\log_e\frac{y}{x}\right|=y^2$ (B) $\left|\log_e\frac{y}{x}\right|=x$ (C) $2\left|\log_e\frac{x}{y}\right|=y+1$ (D) $\left|\log_e\frac{x}{y}\right|=y$

Ans. D

Sol. $y\frac{dx}{dy}=x\left(\ell nx-\ell ny+1\right)$

$$\frac{dx}{dy}=\frac{x}{y}\left(\ell n\frac{x}{y}+1\right)$$

Let $x=vy\Rightarrow\frac{dx}{dy}=v+y\frac{dv}{dy}$

$$\Rightarrow v+y\frac{dv}{dy}=v\ell nv+v\ \Rightarrow y\frac{dv}{dy}=v\ell nv\Rightarrow\int\frac{dv}{v\ell nv}=\int\frac{dy}{y}$$

$$\Rightarrow\ell n\left(\ell nv\right)=\ell ny+\ell nc\Rightarrow\ell nv=yc\Rightarrow\ell n\left(\frac{x}{y}\right)=yc$$

passes through (e, 1) $\Rightarrow\ell ne=c\Rightarrow c=1\Rightarrow y=\ell n\left(\frac{x}{y}\right)\Rightarrow y=\left|\ell n\left(\frac{x}{y}\right)\right|\ \left(\because x>0,y>0\right)$

34. The temperature T(t) of a body at time t = 0 is 160° F and it decreases continuously as per the differential equation $\frac{dT}{dt}=-k\left(T-80\right)$, where K is is a positive constant. If T(15) = 120°F, then T(45) is equal to : **[JEE(Main)Jan.2024]**

एक वस्तु का तापामान T(t), समय t = 0 पर 160° F है तथा यह अवकल समीकरण $\frac{dT}{dt}=-k\left(T-80\right)$ के अनुसार लगातार कम हो j gkg\$t gkjK धनात्मक अचर है। यदि T(15) = 120°F है, तो T(45) बराबर है : **[JEE(Main)Jan.2024]**

(A) 80° F (B) 85° F (C) 95° F (D) 90° F

Ans. D

Sol. $\int\frac{dT}{T-80}=\int-kdt$

$\ell n\,|T-80| = -kt + c$

$At = 0 \quad T = 160^\circ \quad \Rightarrow c = \ell n\,80$

$\ell n\,|T-80| = -kt + \ell n 80$

$\ell n\left|\frac{T-80}{80}\right| = -kt$

$T = 80 + 80e^{-kt}$

when $t = 15$, and $T = 120$ $\Rightarrow k = \frac{1}{15}\ell n\,2$

when $t = 45$, $T = 80 + 80e^{-\ell n 8} = 90^\circ$

35. Let y = y(x) be the solution of the differential equation $\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1$, y(0) = 1. Then, $\left(\frac{1}{\sqrt{2}} + y\left(\frac{1}{\sqrt{2}}\right)\right)^2$ equals: **[JEE(Main)Jan.2024]**

माना अवकलन समीकरण $\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1$, y(0) = 1 का हल y = y(x) है। तो $\left(\frac{1}{\sqrt{2}} + y\left(\frac{1}{\sqrt{2}}\right)\right)^2$ बराबर gS **[JEE(Main)Jan.2024]**

(A) $\frac{4}{4+\sqrt{e}}$ (B) $\frac{3}{3-\sqrt{e}}$ (C) $\frac{1}{2-\sqrt{e}}$ (D) $\frac{2}{1+\sqrt{e}}$

Ans. C

Sol. $\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1$

Put $x + y = t$

$\Rightarrow \frac{dy}{dx} = \frac{dt}{dx} - 1$

$\frac{dt}{dx} - 1 = 2x(t)^3 - xt$

$\Rightarrow \frac{dt}{2t^3 - t} = x\,dx$

$\int \frac{1}{2t^3 - t}\,dt = \int x\,dx$

$\Rightarrow \int \frac{t}{2t^4 - t^2}\,dt = \int x\,dx$

$t^2 = z$

$2t\,dt = dz$

$$\frac{1}{2}\int\frac{dz}{2z^2-z}=\int x dx$$

$$\ell n\left|\frac{z-\frac{1}{2}}{z}\right|=x^2+c$$

$$\ell n\left|\frac{(x+y)^2-\frac{1}{2}}{(x+y)^2}\right|=x^2+c$$

$$y(0)=1\Rightarrow c=\ell n\left(\frac{1}{2}\right)$$

$$\Rightarrow\frac{(x+y)^2-\frac{1}{2}}{(x+y)^2}=e^{x^2}\times\frac{1}{2}$$

$$\Rightarrow\frac{(x+y)^2-\frac{1}{2}}{(x+y)^2}=\sqrt{e}\times\frac{1}{2}$$

$$\Rightarrow(x+y)^2=\frac{1}{2-\sqrt{e}}$$

36. Let α be a non-zero real number. Suppose $f:R\to R$ is a differentiable function such that $f(0)=2$ and $\lim\limits_{x\to-\infty}f(x)=1$. If $f'(x)=\alpha f(x)+3$, for all $x\in R$, then $f(-\log_e 2)$ is equal to : **[JEE(Main)Jan.2024]**

माना α एक शून्येत्तर वास्तविक संख्या है। माना एक अवकलनीय फलन $f:R\to R$ के लिए $f(0)=2$ तथा $\lim\limits_{x\to-\infty}f(x)=1$ हैं। यदि $f'(x)=\alpha f(x)+3$, $x\in R$ है, तो $f(-\log_e 2)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 9 (B) 5 (C) 7 (D) 3

Ans. Bonus

Sol. $f(0)=2,\quad \lim\limits_{x\to-\infty}f(x)=1$

$$f'(x)=\alpha f(x)+3$$

$$\frac{dy}{dx}=\alpha y+3$$

$$\int\frac{dy}{\alpha y+3}=\int dx$$

$$\frac{1}{\alpha}\ell n(\alpha y+3)=x+C$$

$$\alpha y+3=ce^{\alpha x}$$

$$\because \lim_{x\to-\infty} f(x)=1 \qquad \Rightarrow \alpha>0 \text{ but } \alpha=-3$$

So given data contradicts the information

37. If the solution of the differential equation $(2x + 3y - 2)\,dx + (4x + 6y - 7)\,dy = 0$, $y(0) = 3$, is $\alpha x + \beta y + 3\log_e|2x + 3y - \gamma| = 6$, then $\alpha + 2\beta + 3\gamma$ is equal to _____. **[JEE(Main)Jan.2024]**

यदि अवकल समीकरण $(2x + 3y - 2)\,dx + (4x + 6y - 7)\,dy = 0$, $y(0) = 3$ का हल $\alpha x + \beta y + 3\log_e|2x + 3y - \gamma| = 6$ है, तो $\alpha + 2\beta + 3\gamma$ बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 29

Sol. $(2x + 3y - 2)dx + (4x + 6y - 7)dy = 0$

Let $2x+3y=t$

$$2+3\frac{dy}{dx}=\frac{dt}{dx}$$

$$\frac{1}{3}\left[\frac{dt}{dx}-2\right]=-\left(\frac{t-2}{2t-7}\right)$$

$$\frac{dt}{dx}-2=-\left(\frac{3t-6}{2t-7}\right)$$

$$\frac{dt}{dx}=2-\frac{3t-6}{2t-7}$$

$$\frac{dt}{dx}=\frac{t-8}{2t-7}$$

$$\int\left(\frac{2t-7}{t-8}\right)dt=\int dx$$

$$=\int 2dt+\int\frac{a}{t-8}dt=\int dx$$

$$=2t+9\ln|t-8|=x+c$$

$$=2(2x+3y)+9|2x+3y-8|=x+c$$

38. If the solution curve, of the differential equation $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ passing through the point (2, 1) is

$\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{\beta}\log_e\left(\alpha+\left(\frac{y-1}{x-1}\right)^2\right)=\log_e|x-1|$, then $5\beta + \alpha$ is equal to _____.

[JEE(Main)Jan.2024]

यदि अवकलन समीकरण $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ का हल वक्र, जो बिंदु (2, 1) से होकर जाता है,

$\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{\beta}\log_e\left(\alpha+\left(\frac{y-1}{x-1}\right)^2\right)=\log_e|x-1|$ है, तो $5\beta+\alpha$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 11

Sol. $\frac{dy}{dx}=\frac{x+y-2}{x-y}$

$x+y-2=0$ and $x-y=0$ intersect at (1, 1)

$x=X+1, y=Y+1$

$\Rightarrow \frac{dY}{dX}=\frac{X+Y}{X-Y}$

Put $Y=tX \rightarrow \frac{dY}{dX}=t+X\frac{dt}{dX}$

$\Rightarrow t+X\frac{dt}{dX}=\frac{1+t}{1-t}$

$X\frac{dt}{dX}=\frac{1+t^2}{1-t}$

$\frac{dX}{X}=\frac{1-t}{1+t^2}dt$

$\ell n|X|=\tan^{-1}t-\frac{1}{2}\ln|1+t^2|$

$\ell n|x-1|=\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ell n\left|1+\left(\frac{y-1}{x-1}\right)^2\right|+C$

Passes through (2, 1) $\Rightarrow C=0$

$\Rightarrow \tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{2}\ell n\left(1+\left(\frac{y-1}{x-1}\right)^2\right)=\ell n|x-1|$

$\Rightarrow 5\beta+\alpha=11$

39. If the solution curve $y=y(x)$ of the differential equation $(1+y^2)(1+\log_e x)\,dx+x\,dy=0, x>0$ passes through the

point (1, 1) and $y(e)=\frac{\alpha-\tan\left(\frac{3}{2}\right)}{\beta+\tan\left(\frac{3}{2}\right)}$, then $\alpha+2\beta$ is_____. **[JEE(Main)Jan.2024]**

यदि अवकल समीकरण $(1+y^2)(1+\log_e x)\,dx + x\,dy = 0,\ x>0$ का हल वक्र $y = y(x)$, बिन्दु (1, 1) से होकर जाता है तथा

$y(e)=\frac{\alpha-\tan\left(\frac{3}{2}\right)}{\beta+\tan\left(\frac{3}{2}\right)}$ है, तो $\alpha+2\beta$ बराबर है_____. **[JEE(Main)Jan.2024]**

Ans. 3

Sol. $\left(1+y^2\right)(1+\ell n\,x)dx + xdy = 0$

$$\left(\frac{1+\ell n\,x}{x}\right)dx = -\frac{dy}{1+y^2}$$

$$\int\left(\frac{1+\ell n\,x}{x}\right)dx = \int -\frac{dy}{1+y^2}$$

$$\tan^{-1} y = -\frac{(1+\ln x)^2}{2} + C$$

curve passes thraugh $(1, 1) \Rightarrow C = \frac{\pi}{4}+\frac{1}{2}$

$$y = \tan\left(\frac{\pi}{4}+\frac{1}{2}-\frac{(1+\ell n\,x)^2}{2}\right)$$

$$y(e) = \tan\left(\frac{\pi}{4}+\frac{1}{2}-2\right)$$

$$y(e) = \tan\left(\frac{\pi}{4}-\frac{3}{2}\right) = \frac{1-\tan\left(\frac{3}{2}\right)}{1+\tan\left(\frac{3}{2}\right)}$$

Hence $\alpha = \beta = 1$

$\alpha + 2\beta = 3$

40. Let $f(x)=\sqrt{\lim\limits_{r\to x}\left\{\frac{2r^2\left[(f(r))^2-f(x)f(r)\right]}{r^2-x^2}-r^3e^{\frac{f(r)}{r}}\right\}}$ be differentiable in $(-\infty,0)\cup(0,\infty)$ and $f(1)=1$. Then the value of e^a, such that $f(a)=0$, is equal to _____. **[JEE(Main)Jan.2024]**

माना $f(x)=\sqrt{\lim\limits_{r\to x}\left\{\frac{2r^2\left[(f(r))^2-f(x)f(r)\right]}{r^2-x^2}-r^3e^{\frac{f(r)}{r}}\right\}}$ अंतराल $(-\infty,0)\cup(0,\infty)$ में अवकलनीय है तथा $f(1)=1$ है। यदि $f(a)=0$ है, तो e^a का मान है _____. **[JEE(Main)Jan.2024]**

Ans. 2

Sol. $f(x)=\sqrt{\lim\limits_{r\to x}\left\{\frac{2r^2\left[(f(r))^2-f(x)f(r)\right]}{r^2-x^2}-r^3e^{\frac{f(r)}{r}}\right\}}$

$\Rightarrow (f(x))^2=xf(x)f'(x)-x^3e^{\frac{f(x)}{x}}$ by L.H. rule

$\Rightarrow y^2=xy\frac{dy}{dx}-x^3e^{\frac{y}{x}}$

$\Rightarrow \frac{y}{x}=\frac{dy}{dx}-\frac{x^2}{y}e^{\frac{y}{x}}$

Let $y=vx\therefore \frac{dy}{dx}=v+x\frac{dv}{dx}$

$\Rightarrow v=v+x\frac{dv}{dx}-\frac{x^2}{vx}e^v$

$\Rightarrow x\frac{dv}{dx}-\frac{x}{v}e^v=0$

$\Rightarrow ve^{-v}dv-dx=0$

$\Rightarrow (x+c)e^v+v+1=0$ (Integration)

$\Rightarrow (x+c)e^{y/x}+\frac{y}{x}+1=0$

$\therefore y(1)=1\therefore c=-1-\frac{2}{e}$

$\therefore\left(x-1-\frac{2}{e}\right)e^{y/x}+\frac{y}{x}+1=0$

$f(a)=0\Rightarrow x=a,y=0\therefore\left(a-1-\frac{2}{e}\right)+1=0$

$ae - e - 2 + e = 0$

$ae = 2$

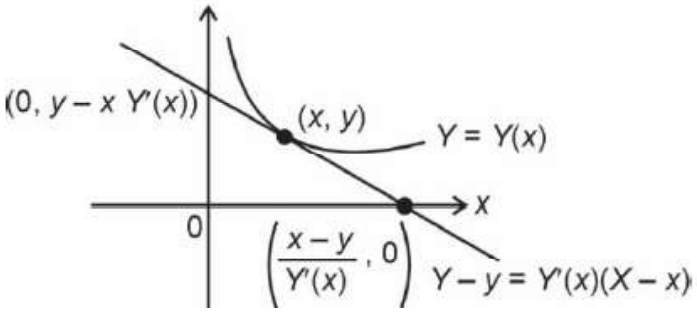
41. Let Y=Y(X) be a curve lying in the first quadrant such that the area enclosed by the line $Y - y = Y'(x)(X - x)$ and the co-ordinate axes, where (x, y) is any point on the curve, is always $\frac{-y^2}{2Y'(x)} + 1$, $Y'(x) \neq 0$. If $Y(1) = 1$, then $12Y(2)$ equals _____. **[JEE(Main)Jan.2024]**

माना प्रथम चतुर्थांश में एक वक्र Y=Y(X) इस प्रकार है कि रेखा $Y - y = Y'(x)(X - x)$ तथा निर्देशांक अक्षों से घिरे क्षेत्र का क्षेत्रफल $\frac{-y^2}{2Y'(x)} + 1$, $Y'(x) \neq 0$ है, (x, y) वक्र पर कोई बिन्दु है। यदि $Y(1) = 1$ है, तो $12Y(2)$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 20

Sol.



$$\Rightarrow \frac{1}{2}\left(y - xY(x)\right)\left(x - \frac{y}{Y(x)}\right) = -\frac{y^2}{2Y(x)} + 1$$

$$\left(y - xY'(x)\right)\left(xY'(x) - y\right) = -y^2 + 2Y'(x)$$

$$yxY'(x) - y^2 - x^2\left(Y'(x)\right) + xyY'(x) = -y^2 + 2Y'(x)$$

$$\left(y - x\frac{dy}{dx}\right)\left(x\frac{dy}{dx} - y\right) = -y^2 + 2\frac{dy}{dx}$$

$$-y^2 - x^2\left(\frac{dy}{dx}\right)^2 + 2xy\left(\frac{dy}{dx}\right) = -y^2 + 2\frac{dy}{dx}$$

$$-x^2\frac{dy}{dx} + 2xy = 2$$

$$\Rightarrow \frac{dy}{dx} + \left(-\frac{2}{x}\right)y = \left(-\frac{2}{x^2}\right)$$

$$\text{I.F.} = \frac{1}{x^2}$$

$\Rightarrow$ Solution is $\frac{y}{x^2}=\frac{-2\cdot x^{-3}}{-3}+c$

$\because y(1)=1$

$\Rightarrow c=\frac{1}{3}$

$\frac{y}{x^2}=\frac{2}{3x^3}+\frac{1}{3}$, when $x=2$, $y=4\left(\frac{1}{12}+\frac{1}{3}\right)=\frac{5}{3}$

$\Rightarrow 12y(2)=20$

42. Let y = y (x) be the solution of the differential equation $\sec^2 x dx+\left(e^{2y}\tan^2 x+\tan x\right)dy=0$, $0<x<\frac{\pi}{2}, y\left(\frac{\pi}{4}\right)=0$. If $y\left(\frac{\pi}{6}\right)=\alpha$, then $e^{8\alpha}$ is equal to_____. **[JEE(Main)Jan.2024]**

माना अवकल समीकरण $\sec^2 x dx+\left(e^{2y}\tan^2 x+\tan x\right)dy=0$, $0<x<\frac{\pi}{2}$, $y\left(\frac{\pi}{4}\right)=0$ का हल y = y(x) है। यदि $y\left(\frac{\pi}{6}\right)=\alpha$ है, तो $e^{8\alpha}$ बराबर है_____ **[JEE(Main)Jan.2024]**

Ans. 9

Sol. $\sec^2 x dx+\left(e^{2y}\tan^2 x+\tan x\right)dy=0$

Put $\tan x=t$

$\sec^2 x dx=dt$

$dt+\left(e^{2y}t^2+t\right)dy=0$

$\frac{1}{t^2}\times\frac{dt}{dy}+\frac{1}{t}=-e^{2y}$

Put $\frac{1}{t}=u$

$-\frac{1}{t^2}\frac{dt}{dy}=\frac{du}{dy}$

$-\frac{du}{dy}+u=-e^{2y}$

$\frac{du}{dy}-u=e^{2y}$

I.f. $=e^{-\int dy}=e^{-y}$

$u\cdot e^{-y}=\int e^{-y}e^{2y}dy$

43. If x= x(t) is the solution of the differential equation $(t+1)dx = (2x + (t+1)^4)dt$, $x(0) = 2$, then $x(1)$ equals _____.
[JEE(Main)Jan.2024]

यदि अवकल समीकरण $(t + 1)dx = (2x + (t + 1)^4)dt$, $x(0) = 2$ का हल $x = x(t)$ है, तो $x(1)$ बराबर है _____
[JEE(Main)Jan.2024]

Ans. 14

Sol. $(t + 1)dx = (2x + (t + 1)^4)dt$

$$\frac{dx}{dt} - \frac{2x}{(t+1)} = (t+1)^3$$

$$IF = e^{-\int \frac{2}{t+1} dt} = \frac{1}{(t+1)^2}$$

$$\frac{x}{(t+1)^2} = \int (t+1)\, dt$$

$$\frac{x}{(t+1)^2} = \frac{t^2}{2} + t + C$$

Now $x(0) = 2$

$\Rightarrow C = 2$

$$\therefore x = \left(\frac{t^2}{2} + t + 2\right)(t+1)^2$$

$$x(1) = \left(\frac{1}{2} + 1 + 2\right)(1+1)^2$$

$$= \frac{7}{2} \times 4 = 14$$

44. If $\frac{dx}{dy} = \frac{1 + x - y^2}{y}$, $x(1) = 1$, then $5x(2)$ is equal to _____. **[JEE(Main)Jan.2024]**

यदि $\frac{dx}{dy} = \frac{1 + x - y^2}{y}$, $x(1) = 1$ है, तो $5x(2)$ बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 5

Sol. $$\frac{dx}{dy} - \frac{x}{y} = \frac{1 - y^2}{y}$$

Integration factor $= e^{\int -\frac{1}{y} dy} = \frac{1}{y}$

$$x \cdot \frac{1}{y} = \int \frac{1-y^2}{y^2} dy$$

$$\frac{x}{y} = \frac{-1}{y} - y + c$$

$$x = -1 - y^2 + cy$$

$$x(1) = 1$$

$$1 = -1 - 1 + c \Rightarrow c = 3$$

$$x = -1 - y^2 + 3y$$

$$5x(2) = 5(-1 - 4 + 6)$$

$$= 5$$

45. If the solution $y = y(x)$ of the differential equation $(x^4 + 2x^3 + 3x^2 + 2x + 2)dy - (2x^2 + 2x + 3)dx = 0$ satisfies $y(-1) = -\frac{\pi}{4}$, then $y(0)$ is equal to : **[JEE(Main)April,2024]**

यदि अवकल समीकरण $(x^4 + 2x^3 + 3x^2 + 2x + 2)dy - (2x^2 + 2x + 3)dx = 0$ का हल $y = y(x)$, $y(-1) = -\frac{\pi}{4}$ को संतुष्ट d j r kgSतो $y(0)$ बराबर है : **[JEE(Main)April,2024]**

(A) 0 (B) $-\frac{\pi}{12}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Ans. D

Sol. $$\int dy = \int \frac{\left(2x^2 + 2x + 3\right)}{x^4 + 2x^3 + 2x + 2} dx$$

$$y = \int \frac{\left(2x^2 + 2x + 3\right)}{\left(x^2 + 1\right)\left(x^2 + 2x + 2\right)} dx$$

$$y = \int \frac{dx}{x^2 + 2x + 2} + \int \frac{dx}{x^2 + 1}$$

$$y = \tan^{-1}(x + 1) + \tan^{-1}x + C$$

$$y(-1) = -\frac{\pi}{4}$$

$$-\frac{\pi}{4} = 0 - \frac{\pi}{4} + C \qquad \Rightarrow C = 0$$

$$\Rightarrow y = \tan^{-1}(x + 1) + \tan^{-1}x$$

$$y(0) = \tan^{-1} 1 = \frac{\pi}{4}$$

46. Let $y = y(x)$ be the solution of the differential equation $(x^2+4)^2dy + (2x^3y + 8xy - 2)dx = 0$. If $y(0) = 0$, then $y(2)$ is equal to : **[JEE(Main)April,2024]**

माना $y = y(x)$ अवकल समीकरण $(x^2 + 4)^2dy + (2x^3y + 8xy - 2)dx = 0$ का हल है। यदि $y(0) = 0$ है, तो $y(2)$ बराबर है : **[JEE(Main)April,2024]**

(A) $\frac{\pi}{32}$ (B) $\frac{\pi}{16}$ (C) 2π (D) $\frac{\pi}{8}$

Ans. A

Sol. $\frac{dy}{dx} + \left(\frac{2x}{x^2+4}\right)y = \frac{2}{\left(x^2+4\right)^2}$

I.F. $= e^{\int \frac{2x}{x^2+4}dx} = e^{\ln\left(x^2+4\right)} = x^2+4$

solution of differential equation $y\left(x^2+4\right) = \int \frac{2}{x^2+4}dx \Rightarrow y\left(x^2+4\right) = \tan^{-1}\left(\frac{x}{2}\right) + C$

But $y(0) = 0 \therefore c = 0$

$y\left(x^2+4\right) = \tan^{-1}\left(\frac{x}{2}\right)$

at $x = 2 \rightarrow y = \frac{\pi/4}{8} = \frac{\pi}{32}$

47. If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} + 2y = \sin(2x)$, $y(0) = \frac{3}{4}$, then $y\left(\frac{\pi}{8}\right)$ is equal to : **[JEE(Main)April,2024]**

यदि अवकल समीकरण $\frac{dy}{dx} + 2y = \sin(2x)$, $y(0) = \frac{3}{4}$ का हल $y = y(x)$ है, तो $y\left(\frac{\pi}{8}\right)$ बरारबर है : **[JEE(Main)April,2024]**

(A) $e^{-\pi/4}$ (B) $e^{-\pi/8}$ (C) $e^{\pi/8}$ (D) $e^{\pi/4}$

Ans. A

Sol. $\frac{dy}{dx} + 2y = \sin 2x$

IF $= e^{\int 2dx} = e^{2x}$

$y.e^{2x} = \int e^{2x} \sin 2x\, dx + c$

$$= \frac{e^{2x}}{8}(2\sin 2x - 2\cos 2x) + c$$

$$y(0) = \frac{3}{4}$$

$$\frac{3}{4} = \frac{1}{8}(-2) + c \Rightarrow c = 1$$

Put $x = \frac{\pi}{8}$

$$y = \frac{1}{8} \times 2\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) + e^{-\pi/4}$$

$$y = e^{-\pi/4}$$

48. The differential equation of the family of circles passing through the origin and having center at the line $y = x$ is :
[JEE(Main)April,2024]

मूल बिन्दु से होकर जाने वाले तथा रेखा $y = x$ पर केन्द्र के वृत्त कुल का अवकल समीकरण है : **[JEE(Main)April,2024]**

(A) $(x^2 - y^2 + 2xy)dx = (x^2 - y^2 - 2xy)dy$ (B) $(x^2 + y^2 + 2xy)dx = (x^2 - y^2 + 2xy)dy$

(C) $(x^2 + y^2 - 2xy)dx = (x^2 + y^2 + 2xy)dy$ (D) $(x^2 + y^2 + 2xy)dx = (x^2 + y^2 - 2xy)dy$

Ans. A

Sol. $C \equiv x^2 + y^2 + gx + gy = 0$ .....(1)

$2x + 2yy' + g + gy' = 0$

$$g = -\left(\frac{2x + 2yy'}{1 + y'}\right)$$

put in (1)

$$x^2 + y^2 - \left(\frac{2x + 2yy'}{1 + y'}\right)(x + y) = 0$$

$$(x^2 - y^2 - 2xy)y' = x^2 - y^2 + 2xy$$

49. Let $y = y(x)$ be the solution of the differential equation $(2x\log_e x)\frac{dy}{dx} + 2y = \frac{3}{x}\log_e x, x > 0$ and $y(e^{-1}) = 0$.
Then, $y(e)$ is equal to : **[JEE(Main)April,2024]**

माना $y = y(x)$ अवकल समीकरण $(2x\log_e x)\frac{dy}{dx} + 2y = \frac{3}{x}\log_e x, x > 0$ तथा $y(e^{-1}) = 0$ का हल है, तो $y(e)$ बराबर है :
[JEE(Main)April,2024]

(A) $-\frac{3}{2e}$ (B) $-\frac{3}{e}$ (C) $-\frac{2}{e}$ (D) $-\frac{2}{3e}$

Ans. B

Sol. $\left(1+x^2\right)\cdot\frac{dy}{dx}+y=e^{\tan^{-1}x}$

Let $\tan^{-1}x=t$

$\frac{1}{1+x^2}\cdot dx=dt$

$\frac{dy}{dt}+y=e^t$

$If=e^{\int 1\cdot dt}=e^t$

$y\cdot e^t=\frac{e^{2t}}{2}+C$

$y\cdot e^{\tan^{-1}x}=\frac{e^{2\tan^{-1}x}}{2}+C$

$x=1 \quad y=0 \quad C=-\frac{e^{\pi/2}}{2}$

$y\cdot\tan^{-1}x=\frac{e^{2\tan^{-1}x}}{2}-\frac{e^{\pi/2}}{2}$

$x=0\Rightarrow y=\frac{1}{2}\left(1-e^{\pi/2}\right)$

50. Let $y=y(x)$ be the solution of the differential equation $\left(1+x^2\right)\frac{dy}{dx}+y=e^{\tan^{-1}x}$, $y(1)=0$. Then $y(0)$ is :

[JEE(Main)April,2024]

माना $y=y(x)$ अवकल समीकरण $\left(1+x^2\right)\frac{dy}{dx}+y=e^{\tan^{-1}x}$, $y(1)=0$ का हल है, तो $y(0)$ है :

[JEE(Main)April,2024]

(A) $\frac{1}{2}\left(1-e^{\pi/2}\right)$ (B) $\frac{1}{4}\left(e^{\pi/2}-1\right)$ (C) $\frac{1}{4}\left(1-e^{\pi/2}\right)$ (D) $\frac{1}{2}\left(e^{\pi/2}-1\right)$

Ans. A

Sol. $\left(1+x^2\right)\cdot\frac{dy}{dx}+y=e^{\tan^{-1}x}$

Let $\tan^{-1}x=t$

$$\frac{1}{1+x^2}\cdot dx = dt$$

$$\frac{dy}{dt} + y = e^t$$

$$\text{If} = e^{\int 1\cdot dt} = e^t$$

$$y\cdot e^t = \frac{e^{2t}}{2} + C$$

$$y\cdot e^{\tan^{-1}x} = \frac{e^{2\tan^{-1}x}}{2} + C$$

$$x = 1 \quad y = 0 \quad C = -\frac{e^{\pi/2}}{2}$$

$$y\cdot \tan^{-1}x = \frac{e^{2\tan^{-1}x}}{2} - \frac{e^{\pi/2}}{2}$$

$$x = 0 \quad \Rightarrow y = \frac{1}{2}\left(1 - e^{\pi/2}\right)$$

51. Suppose the solution of the differential equation $\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - 2\alpha y - (\beta\gamma - 4\alpha)}$ represents a circle passing through origin. Then the radius of this circle is : **[JEE(Main)April,2024]**

मान लीजिए कि अवकल समीकरण $\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - 2\alpha y - (\beta\gamma - 4\alpha)}$ का हल मूल बिन्दु से गुजरने वाले वृत्त को दर्शाता है, तो इस वृत्त d hf=kT; kgS **[JEE(Main)April,2024]**

(A) $\frac{\sqrt{17}}{2}$ (B) 2 (C) $\sqrt{17}$ (D) $\frac{1}{2}$

Ans. A

Sol. $$\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - y(2\alpha+\beta) + 4\alpha}$$

$$\beta x dy - (2\alpha+\beta)ydy + 4\alpha dy = (2+\alpha)xdx - \beta y dx + 2dx$$

$$\beta(xdy + ydx) - (2\alpha+\beta)ydy + 4\alpha dy = (2+\alpha)xdx + 2dx$$

$$\beta xy - \frac{(2\alpha+\beta)y^2}{2} + 4\alpha y = \frac{(2+\alpha)x^2}{2}$$

$\Rightarrow \beta = 0$ for this to be circle

$$\left(2+\alpha\right)\frac{x^2}{2}+\alpha y^2+2x-4\alpha y=0$$

coeff. of $x^2 = y^2$ $\Rightarrow 2 + a = 2a$

$\Rightarrow \alpha = 2$

i.e. $2x^2 + 2y^2 + 2x - 8y = 0$

$x^2 + y^2 + x - 4y = 0$

$$rd = \sqrt{\frac{1}{4}+4} = \frac{\sqrt{17}}{2}$$

52. Let $y = y(x)$ be the solution of the differential equation $(1 + y^2)e^{\tan x}\,dx + \cos^2 x\,(1 + e^{2\tan x})dy = 0$, $y(0) = 1$. Then $y\left(\frac{\pi}{4}\right)$ is equal to : **[JEE(Main)April,2024]**

माना $y = y(x)$ अवकल समीकरण $(1 + y^2)e^{\tan x}\,dx + \cos^2 x\,(1 + e^{2\tan x})dy = 0$ का हल है एवं $y(0) = 1$ है तब $y\left(\frac{\pi}{4}\right)$ बराबर होगा: **[JEE(Main)April,2024]**

(A) $\frac{2}{e}$ (B) $\frac{2}{e^2}$ (C) $\frac{1}{e^2}$ (D) $\frac{1}{e}$

Ans. D

Sol. $(1 + y^2)e^{\tan x}\,dx + \cos^2 x\,(1 + e^{2\tan x})dy = 0$

$$\frac{dy}{1+y^2} = -\frac{e^{\tan x}dx}{\cos^2 x\left(1+e^{2\tan x}\right)}$$

Let $e^{\tan x} = t$

then $e^{\tan x} \cdot \sec^2 x\,dx = dt$

$$\Rightarrow \tan^{-1} y = -\frac{dt}{\left(1+t^2\right)}$$

$\Rightarrow \tan^{-1}y = -\tan^{-1}(t) + C$

$\Rightarrow \tan^{-1}y + \tan^{-1}(e^{\tan x}) = C$

$y(0) = 1$

$\Rightarrow \tan^{-1}(1) + \tan^{-1}(1) = C$

$$C = \frac{\pi}{2}$$

$\rightarrow \tan^{-1} y + \tan^{-1}\left(e^{\tan x}\right) = \frac{\pi}{2}$

$\rightarrow y\left(\frac{\pi}{4}\right) = ?$

$\Rightarrow \tan^{-1}(y) + \tan^{-1}(e) = \frac{\pi}{2}$

$\Rightarrow \tan^{-1}\left(\frac{y+e}{1-ye}\right) = \frac{\pi}{2}$

$\frac{y+e}{1-ye} = \tan\left(\frac{\pi}{2}\right)$

$y = \frac{1}{e}$

53. Let f(x) be a positive function such that the area bounded by y = f(x), y = 0 from x = 0 to x = a > 0 is $e^{-a} + 4a^2 + a - 1$. Then the diffrential equation, whose general solution is $y = c_1 f(x) + c_2$, where c_1 and c_2 are arbitrary constants, is : **[JEE(Main)April,2024]**

माना f(x) एक धनात्मक फलन इस प्रकार है कि y = f(x), y = 0 द्वारा x = 0 से x = a > 0 में परिबद्ध क्षेत्रफल $e^{-a} + 4a^2 + a - 1$ है। तब अवकल समीकरण जिसका सामान्य $y = c_1 f(x) + c_2$ है, जहाँ c_1 तथा c_2 स्वैच्छिक अचर है, होगा : **[JEE(Main)April,2024]**

(A) $\left(8e^x - 1\right)\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

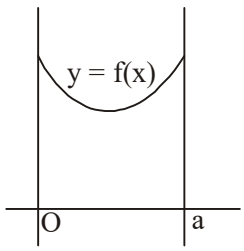
(B) $\left(8e^x + 1\right)\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

(C) $\left(8e^x + 1\right)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

(D) $\left(8e^x - 1\right)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Ans. C

Sol. $\int_0^a f(x)\,dx = e^{-a} + 4a^2 + a - 1$



$f(a) = -e^{-a} + 8a + 1$

$y = C_1\left(1 + 8x - \frac{1}{e^x}\right) + C_2$

$$\frac{dy}{dx} = C_1\left(8 + \frac{1}{e^x}\right) \Rightarrow \frac{\left(\frac{dy}{dx}\right)}{\left(8 + \frac{1}{e^x}\right)} = C_1$$

$$\frac{d^2y}{dx^2}\left(8 + \frac{1}{e^x}\right) - \left(\frac{-1}{e^x}\right)\frac{dy}{dx} = 0$$

54. Let y= y(x) be the solution curve of the differential equation $\sec y \frac{dy}{dx} + 2x \sin y = x^3 \cos y$, $y(1) = 0$. Then $y\left(\sqrt{3}\right)$ is equal to : **[JEE(Main)April,2024]**

माना अवकल समीकरण $\sec y \frac{dy}{dx} + 2x \sin y = x^3 \cos y$, $y(1) = 0$ का हल वक्र $y = y(x)$ है। तो $y\left(\sqrt{3}\right)$ बराबर है : **[JEE(Main)April,2024]**

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{12}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

Ans. C

Sol. $\sec y \frac{dy}{dx} + 2x \sin y = x^3 \cos y$

$\Rightarrow \sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$

Let $z = \tan y$

$\frac{dz}{dx} = \sec^2 y \frac{dy}{dx}$

$\Rightarrow \frac{dz}{dx} + 2xz = x^3$

I.F. $= e^{x^2}$

$\Rightarrow z.e^{x^2} = \int e^{x^2} \cdot x^3 dx + c$

$\Rightarrow \tan y \cdot e^{x^2} = \frac{1}{2}\left(x^2 e^{x^2} - e^{x^2}\right) + c$

$\Rightarrow \tan(0) \cdot e = \frac{1}{2}(1 \cdot e - e) + c$

$\Rightarrow c = 0$

$\Rightarrow \tan y = \frac{x^2 - 1}{2}$

$$f(x)=\tan^{-1}\left(\frac{x^2-1}{2}\right)\Rightarrow f\left(\sqrt{3}\right)=\frac{\pi}{4}$$

55. The solution curve, of the differential equation $2y\frac{dy}{dx}+3=5\frac{dy}{dx}$, passing through the point (0, 1) is a conic, whose vertex lies on the line : **[JEE(Main)April,2024]**

अवकल समीकरण $2y\frac{dy}{dx}+3=5\frac{dy}{dx}$ का हल वक्र, जो (0, 1) से होकर जाता है, एक शांकव है। इस शांकव का शीर्ष निम्न में से किस j§[kkij gS% **[JEE(Main)April,2024]**

(A) $2x+3y=6$ (B) $2x+3y=9$ (C) $2x+3y=-6$ (D) $2x+3y=-9$

Ans. B

Sol. $2y\frac{dy}{dx}+3=5\frac{dy}{dx}$

$2ydy+3dx=5dy$

$y^2+3x=5y+c$, (0, 1)

$1+0=5+c$

$c=-4$

$y^2-5y=-3x-4$

$y^2-5y+\frac{25}{4}-\frac{25}{4}=3x-4$

$\left(y-\frac{5}{2}\right)^2=-3x+\frac{9}{4}$

$\left(y-\frac{5}{2}\right)^2=-3\left(x-\frac{3}{4}\right)$

$\left(\frac{3}{4},\frac{5}{2}\right)$ satisfies by $2x+3y=9$

56. The solution of the differential equation $(x^2+y^2)dx-5xy\,dy=0$, $y(1)=0$, is : **[JEE(Main)April,2024]**

अवकल समीकरण $(x^2+y^2)dx-5xy\,dy=0$, $y(1)=0$ का हल है : **[JEE(Main)April,2024]**

(A) $|x^2-4y^2|^5=x^2$ (B) $|x^2-2y^2|^5=x^2$ (C) $|x^2-4y^2|^6=x$ (D) $|x^2-2y^2|^6=x$

Ans. A

Sol. $(x^2+y^2)dx=5xydy$

$$\frac{dy}{dx}=\frac{x^2+y^2}{5xy}$$

Let $y=vx$

$$\frac{dy}{dx}=v+x\frac{dv}{dx}$$

$$\frac{1+v^2}{5v}=v+x\frac{dv}{dx}$$

$$x\frac{dv}{dx}=\frac{1+v^2}{5v}-v$$

$$x\frac{dv}{dx}=\frac{1-4v^2}{5v}$$

$$\frac{5v}{1-4v^2}dv=\frac{dx}{x}$$

$$\int\frac{dx}{x}=-\frac{5}{8}\int\frac{-8v}{1-4v^2}dv$$

$$\ell nx=-\frac{5}{8}\ell n\left|1-4v^2\right|+\ell nc$$

$$\ell nx+\frac{5}{8}\ell n\left|1-\frac{4y^2}{x^2}\right|=\ell nc \qquad \because\ y(1)=0$$

So $l$nc = 0

$$8\ell nx+5\ell n\left|\frac{x^2-4y^2}{x^2}\right|=0$$

$8lnx + 5ln\,|x^2 - 4y^2| - 10lnx = 0$

$ln\,|x^2 - 4y^2|^5 - lnx^2 = 0$

$|x^2 - 4y^2|^5 = x^2$

57. Let $\int_0^x\sqrt{1-\left(y'(t)\right)^2}dt=\int_0^x y(t)dt$, $0\le x\le 3$, $y\ge 0$, $y(0)=0$. Then at $x=2$, $y''+y+1$ is equal to :

[JEE(Main)April,2024]

माना $\int_0^x\sqrt{1-\left(y'(t)\right)^2}dt=\int_0^x y(t)dt$, $0\le x\le 3$, $y\ge 0$, $y(0)=0$ हैं। तो $x=2$ पर $y''+y+1$ बराबर है :

[JEE(Main)April,2024]

(A) 1 (B) $\sqrt{2}$ (C) 2 (D) $\frac{1}{2}$

Ans. A

Sol. $\sqrt{1-\left(y'(x)\right)^2} = y(x)$

$$1-\left(\frac{dy}{dx}\right)^2 = y^2$$

$$\left(\frac{dy}{dx}\right)^2 = 1-y^2$$

$$\frac{dy}{\sqrt{1-y^2}} = dx \text{ OR } \frac{dy}{\sqrt{1-y^2}} = -dx$$

$\Rightarrow \sin^{-1}y = x + c, \sin^{-1}y = -x + c$

$x = 0, y = 0 \quad \Rightarrow c = 0$

$\sin^{-1}y = x$, as $y \geq 0$

$\sin x = y$

$$\Rightarrow \frac{dy}{dx} = \cos x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\sin x$$

$\Rightarrow -\sin x + \sin x + 1 = 1$

58. Let the solution $y = y(x)$ of the differential equation $\frac{dy}{dx} - y = 1 + 4\sin x$ satisfy $y(\pi) = 1$. Then $y\left(\frac{\pi}{2}\right) + 10$ is equal to _____. **[JEE(Main)April,2024]**

माना अवकल समीकरण $\frac{dy}{dx} - y = 1 + 4\sin x$ के हल $y = y(x)$ के लिए $y(\pi) = 1$ है। तो $y\left(\frac{\pi}{2}\right) + 10$ बराबर है _____

[JEE(Main)April,2024]

Ans. 7

Sol. $\frac{dy}{dx} - y = 1 + 4\sin x$ linear differential equation

I.F. $= e^{\int -dx} = e^{-x}$

So solution of linear differential equation is

$$ye^{-x} = \int e^{-x}(1 + 4\sin x)dx$$

$$ye^{-x} = -e^{-x} + 4\int e^{-x}\sin x dx$$

$$ye^{-x} = -e^{-x} + 4\frac{e^{-x}}{2}\left[-\sin x - \cos x\right] + c$$

$$y = -1 - 2\left(\sin x + \cos x\right) + ce^{x}$$

Now $y(\pi) = 1$

$\Rightarrow 1 = -1 - 2(0 - 1) + ce^{\pi}$

Hence $y = -1 - 2(\sin x + \cos x)$

$$\Rightarrow y\left(\frac{\pi}{2}\right) + 10 = -1 - 2(1 + 0) + 10$$

$= 7$

59. Let $y = y(x)$ be the solution of the differential equation $(x + y + 2)^2\, dx = dy$; $y(0) = -2$. Let the maximum and minimum values of the function $y = y(x)$ in $\left[0, \frac{\pi}{3}\right]$ be α and β, respectively. If $(3\alpha + \pi)^2 + \beta^2 = \gamma + \delta\sqrt{3}$, $\gamma, \delta \in Z$, then $\gamma + \delta$ equals _____. **[JEE(Main)April,2024]**

माना $y = y(x)$ अवकल समीकरण $(x + y + 2)^2\, dx = dy$; $y(0) = -2$ का एक हल है। माना $\left[0, \frac{\pi}{3}\right]$ में फलन $y = y(x)$ के उच्चतम r Hkkfu Eur e eku Øe'k% α तथा β हैं। यदि $(3\alpha + \pi)^2 + \beta^2 = \gamma + \delta\sqrt{3}$, $\gamma, \delta \in Z$ तो $\gamma + \delta$ बराबर है _____

[JEE(Main)April,2024]

Ans. 31

Sol. $\frac{dy}{dx} = (x + y + 2)^2$

Put $x + y + 2 = t$

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - 1 = t^2$$

$\tan^{-1} t = x + c$

$\tan^{-1}(x + y + z) = x + c$

$\Rightarrow (x = 0, y = -2)$

$C = 0$

$y = \tan x - x - 2$

$$\frac{dy}{dx} = \tan^2 x \geq 0$$

$\forall x \in \left[0, \frac{\pi}{3}\right]$

$\Rightarrow y_{min} = y(0) = -2 = \beta$

And $y_{max} = y\left(\frac{\pi}{3}\right) = \sqrt{3} - 2 - \frac{\pi}{3} = \alpha$

Now, $(3\alpha + \pi)^2 + \beta^2$

$\left(3\sqrt{3} - 6 - \pi + \pi\right)^2 + 4$

$27 + 36 - 36\sqrt{3} + 4$

$67 - 36\sqrt{3}$

So, $\gamma = 67$

$\delta = -36$

$\Rightarrow \gamma + \delta = 31$

60. Let f be a differentiable function in the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim\limits_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ for each $x > 0$. Then $2 f(2) + 3 f(3)$ is equal to _____. **[JEE(Main)April,2024]**

माना अंतराल $(0, \infty)$ में एक अवकलनीय फलन f है, जिसके लिए $f(1) = 1$ है तथा प्रत्येक $x > 0$ के लिए $\lim\limits_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ है। तो $2 f(2) + 3 f(3)$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 24

Sol. $f(1) = 1$

$\lim\limits_{x \to \infty} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$

$\lim\limits_{x \to \infty} \frac{2 + f(x) - x^2 f'(t)}{1} = 1$

$= 2xy - x^2 \frac{dy}{dx} = 1$

$= \frac{dy}{dx} - \frac{2y}{x} = \frac{-1}{x^2}$

I.f. $e^{-\int \frac{2}{x} dx} = e^{-2\ell nx} = \frac{1}{x^2}$

$y \cdot \frac{1}{x^2} = \int \frac{-1}{x^4} dx$

$$\frac{y}{x^2}=\frac{-x^{-4+1}}{-4+1}+C$$

$$\frac{y}{x^2}=\frac{1}{3x^3}+C$$

$\because f(1)=1$

$$\frac{1}{1}=\frac{1}{3}+C \Rightarrow C=1-\frac{1}{3}=\frac{2}{3}$$

$$y=\frac{1}{3x}+\frac{2}{3}x^2$$

Now $2f(2)+3f(3)=2\times\left(\frac{1}{6}+\frac{8}{3}\right)+3\times\left(\frac{1}{9}+6\right)$

$$=2\times\left(\frac{1+16}{6}\right)+3\times\left(\frac{54+1}{9/3}\right)=\frac{17}{3}+\frac{55}{3}=\frac{72}{3}=24$$ Ans.

61. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx}+\frac{2x}{\left(1+x^2\right)^2}y=xe^{\frac{1}{\left(1+x^2\right)}}$; $y(0)=0$. Then the area enclosed by the curve $f(x)=y(x)e^{\frac{1}{\left(1+x^2\right)}}$ and the line $y-x=4$ is_____. **[JEE(Main)April,2024]**

माना अवकल समीकरण $\frac{dy}{dx}+\frac{2x}{\left(1+x^2\right)^2}y=xe^{\frac{1}{\left(1+x^2\right)}}$; $y(0)=0$ का हल $y = y(x)$ है। तो वक्र $f(x)=y(x)e^{\frac{1}{\left(1+x^2\right)}}$ तथा रेखा $y - x = 4$ से घिरे क्षेत्र का क्षेत्रफल है_____ **[JEE(Main)April,2024]**

Ans. 18

Sol. $\frac{dy}{dx}+\frac{2x}{\left(1+x^2\right)^2}y=xe^{\frac{1}{1+x^2}}$; $y(0)=0$

I.F. of linear differential equation,

$$\text{I.F.}=e^{\int\frac{2x}{\left(1+x^2\right)^2}dx}=e^{\left(\frac{-1}{1+x^2}\right)}$$

$$\Rightarrow y\left(e^{\left(\frac{-1}{1+x^2}\right)}\right)=\int x\cdot e^{\frac{1}{1+x^2}}\cdot e^{\left(\frac{-1}{1+x^2}\right)}dx$$

$$=\frac{x^2}{2}+c$$

$\Rightarrow y(0) = 0 \Rightarrow 0(e^{-1}) = c \Rightarrow c = 0$

$$\Rightarrow y = \frac{e^{\frac{1}{1+x^2}} \cdot x^2}{2}$$

Area between curve $ye^{\left(\frac{-1}{1+x^2}\right)} = \frac{x^2}{2}$ and $y - x = 4$

$\Rightarrow 2(x + 4) = x^2 \Rightarrow x^2 - 2x - 8 = 0$

$\Rightarrow (x - 4)(x + 2) = 0$

$$\int_{-2}^{4}\left[(x+4)-\frac{x^2}{2}\right]dx = \frac{x^2}{2} + 4x - \frac{x^3}{6}\Bigg|_{-2}^{4}$$

$$= \left(8+16-\frac{64}{6}\right) - \left(2-8+\frac{8}{6}\right)$$

$= 30 - 12 = 18$

62. Let a conic C pass through the point (4, –2) and P(x, y), $x \geq 3$, be any point on C. Let the slope of the line touching the conic C only at a single point P be half the slope of the line joining the points P and (3, –5). If the focal distance of the point (7, 1) on C is d, then 12d equals _____. **[JEE(Main)April,2024]**

माना कोई शंकु C बिन्दु (4, –2) से होकर जाता है तथा C पर कोई बिन्दु P(x, y), $x \geq 3$ है। माना शंकु C को मात्र एक बिन्दु P पर Li 'kadq j usokyh js[kkd hi ø.k kfcUnq P तथा (3, –5) को मिलाने वाली रेखा की प्रवणता की आधी है। यदि C पर स्थित बिन्दु (7, 1) की नाभीय दूरी d है, तो 12d बराबर है _____ **[JEE(Main)April,2024]**

Ans. 75

Sol. P(x, y) and $x \geq 3$

Slope of line at P(x, y) will be $\frac{dy}{dx} = \frac{1}{2}\left(\frac{y+5}{x-3}\right)$

$$\Rightarrow 2\frac{dy}{(y+5)} = \frac{1}{(x-3)}dx$$

$\Rightarrow 2\ell n(y+5) = \ell n(x-3) + C$

Passes through (4, –2)

$\Rightarrow 2\ell n(3) = \ell n(1) + C$

$\Rightarrow C = 2\ell n(3)$

$\Rightarrow 2\ell n(y+5) = \ell n(x-3) + 2\ell n(3)$

$$\Rightarrow 2\left(\ln\left(\frac{y+5}{3}\right)\right)=\ell n\left(x-3\right)$$

$$\Rightarrow\left(\frac{y+5}{3}\right)^2=\left(x-3\right)$$

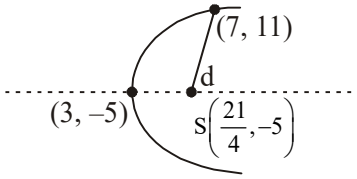
$$\Rightarrow\left(y+5\right)^2=9\left(x-3\right)$$

$$\downarrow$$

Parabola

$4a = 9$

$$a=\frac{9}{4}$$



$$d=\sqrt{\left(\frac{7}{4}\right)^2+6^2}$$

$$d=\frac{\sqrt{625}}{4}$$

$$d=\frac{25}{4}$$

$12d = 75$

63. If the solution y(x) of the given differential equation $(e^y + 1)\cos x dx + e^y \sin x\, dy = 0$ passes through the point $\left(\frac{\pi}{2}, 0\right)$, then the value of $e^{y\left(\frac{\pi}{6}\right)}$ is equal to _____. **[JEE(Main)April,2024]**

यदि अवकल समीकरण $(e^y + 1)\cos x dx + e^y \sin x\, dy = 0$ का हल y(x) बिन्दु $\left(\frac{\pi}{2}, 0\right)$ से गुजरता है, तो $e^{y\left(\frac{\pi}{6}\right)}$ का मान बराबर gS_____ **[JEE(Main)April,2024]**

Ans. 3

Sol. $(e^y + 1)\cos x\, dx + e^y \sin x dy = 0$

$$\cot x dx+\frac{e^y}{e^y+1}dy=0$$

$$\int\cot x dx+\int\frac{e^y}{e^y+1}dy=0$$

$\ln|\sin x| + \ln|e^y + 1| = \ln c$

$|\sin x\,(e^y + 1)| = c$

Put & $x = \frac{\pi}{2}$ $y = 0$

$c = 2$

$\sin x\,(e^y + 1) = 2$

$e^y + 1 = 2\,\text{cosec}\,x$

$e^y = 2\,\text{cosec}\,x - 1$

$e^{y\left(\frac{\pi}{6}\right)} = 2\,\text{cosec}\,\frac{\pi}{6} - 1$

$= 2 \times 2 - 3 = 3$

64. Let $\alpha|x| = |y|e^{xy-\beta}$, $\alpha, \beta \in N$ be the solution of the differential equation $xdy - ydx + xy(xdy + ydx) = 0$, $y(1) = 2$. Then $\alpha + \beta$ is equal to _____. **[JEE(Main)April,2024]**

माना अवकल समीकरण $xdy - ydx + xy(xdy + ydx) = 0$, $y(1) = 2$ का हल $\alpha|x| = |y|e^{xy-\beta}$, $\alpha, \beta \in N$ है। तो $\alpha + \beta$ बराबर gS_____. **[JEE(Main)April,2024]**

Ans. 4

Sol. $\alpha|x| = |y|\,e^{yx-\beta}$, $\alpha, \beta \in N$

$xdy - ydx + xy(xdy + ydx) = 0$

$\frac{dy}{y} - \frac{dx}{x} + (xdy + ydx) = 0$

$\ell n|y| - \ell n|x| + xy = c$

$y(1) = 2$

$\ell n|2| - 0 + 2 = c$

$c = 2 + \ell n2$

$\ell n|y| - \ell n|x| + xy = 2 + \ell n2$

$\ell n|x| = \ell n\left|\frac{y}{2}\right| - 2 + xy$

$|x| = \left|\frac{y}{2}\right|e^{xy-2}$

$2|x| = |y|e^{xy-2}$

$\alpha = 2$ $\beta = 2$ $\alpha + \beta = 4$

65. For a differentiable function $f: R \to R$, suppose $f'(x) = 3f(x) + \alpha$, where $\alpha \in R$, $f(0) = 1$ and $\lim_{x\to-\infty} f(x) = 7$.

Then $9f\left(-\log_e 3\right)$ is equal to _____. **[JEE(Main)April,2024]**

एक अवकलनीय फलन $f : R \to R$ के लिए, माना $f'(x) = 3f(x) + \alpha$, जहाँ $\alpha \in R$, $f(0) = 1$ तथा $\lim\limits_{x\to-\infty} f(x) = 7$ हैं। तो $9f\left(-\log_e 3\right)$ बराबर है _____. **[JEE(Main)April,2024]**

Ans. 61

Sol. $f'(x) = 3f(x) + \alpha$

$$\Rightarrow \frac{dy}{3y+\alpha} = dx$$

$$\Rightarrow \frac{1}{3}\ell n\left(3y+\alpha\right) = x + C$$

$$y(0) = 1 \Rightarrow C = \frac{1}{3}\ell n\left(3+\alpha\right)$$

$$\frac{1}{3}\ell n\left(\frac{3y+\alpha}{3+\alpha}\right) = x$$

$$\Rightarrow y = \frac{1}{3}\left(\left(3+\alpha\right)e^{3x} - \alpha\right) = f(x)$$

$$\lim_{x\to-\infty} f\left(x\right) = 7 \Rightarrow \alpha = -21$$

$$\Rightarrow f\left(x\right) = 7 - 6e^{3x}$$

$$9f\left(-\ell n3\right) = 61$$